



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

BE - CS

III to VIII Semester

Dec.2024/Jan.2025

2021, 2022 Scheme

VVIET

Bannur road, Alanahalli, Mysore-570028

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematical Foundations for Computing, Probability and Statistics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define tautology. Determine whether the following compound statement is a tautology or not.

$$[(p \vee q) \rightarrow r] \leftrightarrow [\sim r \rightarrow \sim (p \vee q)]$$

(06 Marks)
- b. Prove the following using the laws of logic

$$[\sim p \wedge (\sim q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)] \leftrightarrow r$$

(07 Marks)
- c. Give direct proof and proof by contradiction for the statement “If n is an odd integer then $n + q$ is an even integer”.

(07 Marks)

OR

- 2 a. Define : i) Open Statement ii) Quantifiers.

(06 Marks)
- b. Test the validity of the arguments using rules of inference

$$\begin{array}{l} p \rightarrow (q \rightarrow r) \\ p \vee \sim s \\ q \\ \hline \therefore s \rightarrow r \end{array}$$

(07 Marks)
- c. If $p(x) : x \geq 0$, $q(x) : x^2 \geq 0$, $r(x) : x^2 - 3x - 4 = 0$, $s(x) : x^2 - 3 > 0$. Determine the truth or falsity of the following statement:
 - i) $\exists x [p(x) \wedge q(x)]$
 - ii) $\forall x [p(x) \rightarrow q(x)]$
 - iii) $\forall x [q(x) \rightarrow s(x)]$
 - iv) $\forall x [r(x) \wedge s(x)]$
 - v) $\exists x [p(x) \wedge r(x)]$
 - vi) $\forall x [r(x) \rightarrow p(x)]$
 - vii) $\exists x [r(x) \rightarrow \sim p(x)]$

(07 Marks)

Module-2

- 3 a. Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$ if $(gof)(x) = 9x^2 - 9x + 3$. Determine a and b .

(06 Marks)
- b. Let $A = \{1, 2, 3, 4\}$ and let R be the relation on A defined by xRy if and only if “ x divides y ” write down the relation R , relation matrix M_R and draw its diagram.

(07 Marks)
- c. Prove that in every graph the number of vertices of odd degree is even.

(07 Marks)

OR

- 4 a. Draw the Hasse diagram of the relation R on $A = \{1, 2, 3, 4, 5\}$ whose matrix is as given below

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

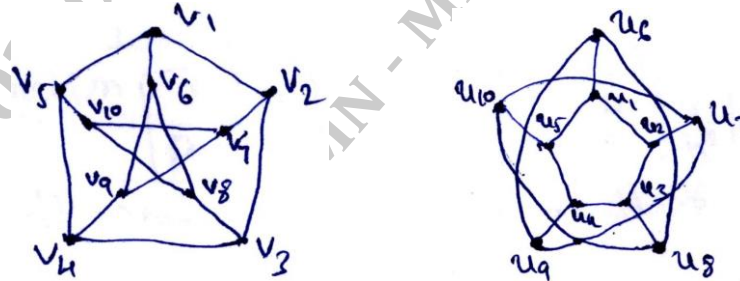
(06 Marks)

- b. Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 5$. Let a function $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = \frac{1}{2}(x - 5)$. Prove that g is an inverse of f .

(07 Marks)

- c. Define graph isomorphism. Determine whether the following graphs are isomorphic or not.

(07 Marks)

**Module-3**

- 5 a. Calculate the coefficient of correlation and obtain the lines of regression for the following data:

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

(06 Marks)

- b. Fit a straight line in the least square sense for the following data:

x	5	10	15	20	25
y	16	19	23	26	30

(07 Marks)

- c. Fit a curve $y = ax^b$ for the following data:

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

(07 Marks)

OR

- 6 a. The following are the percentage of marks in mathematics (x) and statistics (y) of nine students. Calculate the rank correlation coefficient.

x	38	50	42	61	43	55	67	46	72
y	41	64	70	75	44	55	62	56	60

(06 Marks)

- b. Fit a parabola $y = ax^2 + bx + c$ for the data

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1

(07 Marks)

- c. With usual notation, compute means $\bar{x}$, $\bar{y}$ and correlation coefficient r from the following lines of regression, $2x + 3y + 1 = 0$, $x + 6y - 4 = 0$.

(07 Marks)

Module-4

- 7 a. A random variable X has the following probability function:

x	0	1	2	3	4	5	6	7
p(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

Find k , and evaluate $p(x < 6)$, $p(x \geq 6)$, $p(0 < x < 5)$

(06 Marks)

- b. Find the mean and S.D of binomial distribution.

(07 Marks)

- c. The marks of 1000 students in an examination follows a normal distribution with mean 70 and S.D. 5. Find the number of students whose marks will be i) less than 65 ii) more than 75 iii) between 65 and 75 ($\phi(1) = 0.3413$).

(07 Marks)

OR

- 8 a. Find the constant for such that $f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$ is a p.d.f. Also compute $p(1 < x < 2)$, $p(x \leq 1)$, $p(x > 1)$.

(06 Marks)

- b. A shop has 4 diesel generator sets which it hires every day. The demand for a genset on an average is a poisson variate with value $5/2$. Obtain the probability that on a particular day i) there was no demand ii) A demand had to be refused.

(07 Marks)

- c. In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find the mean and S.D of the distribution: ($p(0.5) = 0.19$, $p(1.4) = 0.42$)

(07 Marks)

Module-5

- 9 a. The joint distribution of two random variables x and y is as follows:

x/y	3	4	5
2	1/6	1/6	1/6
5	1/12	1/12	1/12
7	1/12	1/12	1/12

Compute i) $E(x)$ and $E(y)$ variables?

ii) $E(xy)$

iii) $\text{Cov}(x, y)$. Are they independent random

(06 Marks)

- b. A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% level of significance. (07 Marks)
- c. In experiments on pea breeding, the following frequencies of seeds were obtained:

Round and Yellow	Wrinkled and Yellow	Round and Green	Wrinkled and Green	Total
315	101	108	32	556

Theory predicts that the frequencies should be in proportions 9:3:3:1. Examine the correspondence between theory and experiment ($\chi^2_{0.05} = 7.815$ for 3df) (07 Marks)

OR

- 10 a. Explain the terms:
- Null hypothesis
 - Significance level
 - Type I and Type II errors
- (06 Marks)
- b. The mean life of 100 fluorescent tube lights manufactured by a company is found to be 1570 hrs with a S.D of 120 hrs. Test the hypothesis that the mean life time of the lights produced by the company is 1600 hrs at 0.01 level of significance. (07 Marks)
- c. The nine items of a sample have the following values: 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed mean of 47.5? ($t_{0.05} = 2.31$ for 8 d.f). (07 Marks)

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Automata Theory and Compiler Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

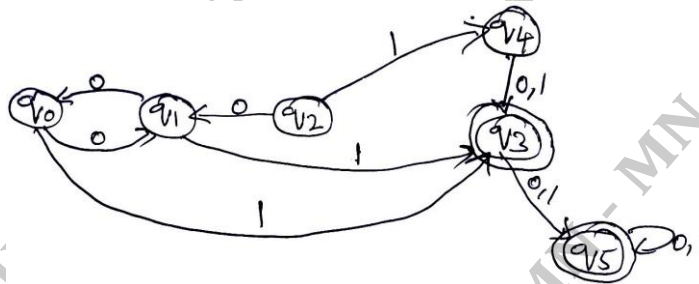
Module-1

- 1 a. Define the three basic concepts of Automata. Also construct a DFA that accepts all strings that have the first and last letter different on $\Sigma = \{a, b\}$. Justify the DFA with example. (10 Marks)
- b. Solve by converting the following NFA to DFA. (10 Marks)



OR

- 2 a. Explain the different phases of a compiler with neat block diagram and convert the source code. Position = Initial + rate * 60 into target code. (10 Marks)
- b. Solve by Minimizing the following DFA. (10 Marks)



Module-2

- 3 a. Define the formal definition of Regular expression. Also write the regular expression for the following : i) Set of strings consisting of Even numbers of 'a' s followed by odd number of 'b' s on $\Sigma = \{a, b\}$.
 ii) $L = \{a^n b^m : (n + m) \text{ is even}\}$.
 iii) $L = \{a^n b^m : n \geq 4, m \leq 3\}$. Justify the answer. (10 Marks)
- b. Explain Input buffering in Lexical Analyzer. Define Token , Patterns and Lexemes with examples. Also write the tokens for $E = m * c * * 2$. (10 Marks)

OR

- 4 a. Define Regular Definitions. Write the Regular Definitions for 'C' identifiers and unsigned numbers using short hands notations and write the transition diagram. (10 Marks)
- b. State and prove pumping lemma theorem for Regular languages. (10 Marks)

Module-3

- 5 a. Define Context free grammar. Write a CFG for the following :
 i) To generate strings of palindrome over $\Sigma = \{0, 1\}$.
 ii) $L = \{a^i b^j \mid i \neq j, i \geq 0 \text{ and } j \geq 0\}$
 iii) $L = \{0^m 1^n 2^n \mid m \geq 1, n \geq 0\}$. Justify the answer. (10 Marks)

- b. Define Left recursion and left factoring. Also remove the left recursion and left factoring for the Grammar
- $$E \rightarrow E + T \mid T$$
- $$T \rightarrow id \mid id [] \mid id [X]$$
- $$X \rightarrow E, E \mid E.$$

(10 Marks)

OR

- 6 a. Define Ambiguous grammar. Show that the following is ambiguous. (10 Marks)

$S \rightarrow i c t s \mid i c t s e s \mid a$
 $G \rightarrow b$ for the string ibtibtaea

- b. Consider the grammar

$$E \rightarrow T E'$$

$$E' \rightarrow + T E' \mid \epsilon$$

$$T \rightarrow F T'$$

$$T' \rightarrow * F T' \mid \epsilon$$

$$F \rightarrow (E) \mid id$$

- i) Compute FIRST and Follow sets.
 ii) Using FIRST and Follow sets construct the Predictive LL (1) parsing table. (10 Marks)

Module-4

- 7 a. Define Non Deterministic Pushdown Automata. Construct an NPDA for the Language $L = \{W \in (a, b)^* : n_a(w) = n_b(w)\}$ and draw the transition diagram. (10 Marks)
- b. Define Handle and Handle Pruning. For the following grammar perform shift reduce for the string $id_1 + id_2 * id_3$.
- $$E \rightarrow E + E$$
- $$E \rightarrow E * E$$
- $$E \rightarrow (E)$$
- $$E \rightarrow id.$$

(10 Marks)

OR

- 8 a. Define Instantaneous Description in Pushdown Automata. Construct an NPDA for the Language $L = \{WCW^R : W \in (a, b)^*\}$. (10 Marks)
- b. Consider the Grammar.
- $$S \rightarrow L = R \mid R$$
- $$L \rightarrow * R \mid id$$
- $$R \rightarrow L$$

Verify the grammar is SLR (1) or not through the suitable parsing table. (10 Marks)

Module-5

- 9 a. Define Turing Machine. Construct a Turing Machine to recognize the Language. $L = \{a^n b^n : W \in \{a, b\}^* n \geq 1\}$. (10 Marks)
- b. Write the SDD for the grammar. Also construct the Annotated Parse tree for $5 * 6 + 7$;
- $$S \rightarrow EN$$
- $$E \rightarrow E + T$$
- $$E \rightarrow E - T$$
- $$E \rightarrow T$$
- $$T \rightarrow T * F$$
- $$T \rightarrow T / F$$
- $$T \rightarrow F$$
- $$F \rightarrow (E)$$
- $$F \rightarrow digit$$
- $$N \rightarrow ;$$

(10 Marks)

OR

- 10 a. Construct a Turing Machine to recognize the Language.
 $L = \{0^n 1^n 2^n \mid n \geq 1\}$ and trace the string 0 0 1 1 2 2. (12 Marks)
- b. For the Grammar construct the SDD and the annotated parse tree for the string $3 * 5 * 4$ and show the Evaluation order.
- $T \rightarrow FT'$
 $T' \rightarrow * FT'$
 $T' \rightarrow \epsilon$
 $F \rightarrow \text{digit}.$
- (08 Marks)

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21CS52

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss OSI Reference model with a neat diagram. (10 Marks)
- b. Explain Unicast, Multicast and broadcast in computer networks. (10 Marks)

OR

- 2 a. Discuss the following transmission medium with diagram,
 - (i) Co-axial cable.
 - (ii) Fibre optic cable.(10 Marks)
- b. Explain the design issues in computer networks. (10 Marks)

Module-2

- 3 a. Explain error detecting codes and obtain the CRC code for the frame given polynomial 1101011111 using the generator $G(x) = x^4 + x + 1$. (10 Marks)
- b. Explain framing with Byte Count and Flag bits with bit stuffing. (10 Marks)

OR

- 4 a. Illustrate the Hamming code method with an example. (10 Marks)
- b. Explain the following with examples:
 - (i) Binary convolution code.
 - (ii) Reed Solomen code.(10 Marks)

Module-3

- 5 a. Discuss Store and Forward packet switching. (10 Marks)
- b. Explain the services provided by Network layer to Transport layer. (10 Marks)

OR

- 6 a. Discuss shortest path algorithm. (10 Marks)
- b. Explain the approaches to congestion control in Network layer. (10 Marks)

Module-4

- 7 a. Explain Berkeley Sockets in detail. (10 Marks)
- b. Explain socket programming with an example. (10 Marks)

OR

- 8 a. Explain TCP protocol with TCP segment header. (10 Marks)
b. Explain TCP connection establishment and TCP connection release with code snippet. (10 Marks)

Module-5

- 9 a. Explain the process communication in the Application layer. (10 Marks)
b. Discuss the Transport services provided by the Internet. (10 Marks)

OR

- 10 a. Explain the web and HTTP with Request response behaviour. (10 Marks)
b. Discuss the Electronic Mail in the Internet. (10 Marks)

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21CS53

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Database Management Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain in detail the characteristics of database approach. (08 Marks)
- b. Define the following terms and also give example :
i) Database ii) DBMS (04 Marks)
- c. List and explain the advantages of using DBMS Approach. (08 Marks)

OR

- 2 a. Explain cardinality ratio and participation constraints along with an example. (06 Marks)
- b. With a neat diagram explain the three schema architecture. (06 Marks)
- c. Draw an ER diagram for library database by considering at least 5 entities. (08 Marks)

Module-2

- 3 a. Explain in detail characteristics of Relations. (06 Marks)
- b. Discuss different types of update operations on relational database. Also give an example. (06 Marks)
- c. Write a note on Natural join and division operation. (08 Marks)

OR

- 4 a. Consider the 2 tables. Show the result of the following :

R ₁		
a ₁	a ₂	a ₃
20	L	15
15	m	18
25	L	16

R ₂		
b ₁	b ₂	b ₃
20	L	6
25	n	8
28	1	4

(i) $R_1 \bowtie R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

(ii) $R_1 \Join R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

(iii) $R_1 \bowtie R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

(iv) $R_1 \Join R_2$

$(R_1 \cdot a_1 = R_2 \cdot b_1)$

- b. With an example explain steps of ER to Relational Mapping algorithm.

(08 Marks)

(12 Marks)

Module-3

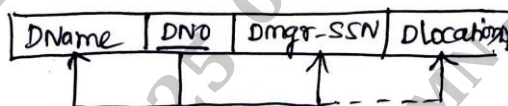
- 5 a. For the following Database schema.
 Employee (Fname, Minit, Lname, SSN, Bdate, Address, Salary, SuperSSN, DNo)
 Department(DName, Dno, Mgr_SSN, Mgr_Startdate)
 Dept_Locations(Dno, Dlocation)
 Project(PName, Prj_no, Plocation, Dnum)
 Works_on(ESSN, Prj_no, Hours)
 Dependent(ESSN, DependentName, Sex, Bdate, Relationship)
 Write SQL Queries for the following :
 (i) Find sum_of_salaries of all employees who work in Dept No 10, average salaries of all employees who work in Dept No 10.
 (ii) List all employees who do not have any dependent.
 (iii) For each project, retrieve the project number and the number of employees who work on that project.
 (iv) Make list of all project numbers for projects that involve an employee whose last name is 'Kumar'. (08 Marks)
- b. Write command that is used for table creation. Explain how primary key, foreign key are specified in SQL during table creation with suitable example. (06 Marks)
- c. Explain view in SQL, with suitable example. (06 Marks)

OR

- 6 a. Explain stored procedures in SQL with example. (06 Marks)
- b. How triggers are defined in SQL? Explain with an example. (06 Marks)
- c. Write a note on : (i) Cursor (ii) Assertions (08 Marks)

Module-4

- 7 a. List and explain the informal Design guidelines for relation schemas. (08 Marks)
- b. Define the following :
 (i) Functional dependency (ii) Key (iii) Superkey (iv) Prime attribute (06 Marks)
- c. For the given schema, discuss the 3 main techniques to achieve first normal form.



(06 Marks)

OR

- 8 a. Explain in detail 2nd Normal form and 3rd Normal form along with example. (08 Marks)
- b. Write an algorithm for determining X^+ , the closure of X under F. Give an example. (06 Marks)
- c. Write a note on 4th Normal form. (06 Marks)

Module-5

- 9 a. Define Transaction. Discuss ACID properties. (06 Marks)
- b. With neat diagram explain transition diagram of a transaction. (06 Marks)
- c. Explain the Lost Update problem and Temporary update problem with respect to concurrent transaction execution. (08 Marks)

OR

- 10 a. Briefly discuss 2-phase locking techniques for concurrency control. (10 Marks)
- b. Write a note on :
 i) Deadlock prevention protocols ii) Basic Timestamp ordering algorithm (10 Marks)

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Research Methodology and Intellectual Property Rights

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Give various definitions of research explaining its meaning and scope. Explain the objective and motivation in Engineering research. (08 Marks)
- b. Describe the steps involved in research process with neat diagram. (06 Marks)
- c. What is ethics in Engineering research? Why it is important. (06 Marks)

OR

- 2 a. What is research problem? Define the main issues which should receive the attention of the research in formulating the research problem. (08 Marks)
- b. Discuss in detail various types of research. (06 Marks)
- c. Describe different types of research misconduct. (06 Marks)

Module-2

- 3 a. Discuss the importance of critical literature review and its uses in planning innovation research. (06 Marks)
- b. Explain the concept of knowledge flow through citation. How collaborations certainly impact citation counts? Explain. (08 Marks)
- c. Write short notes on : (06 Marks)
 - i) Google and Google scholar
 - ii) Acknowledgment and Attributions

OR

- 4 a. Analyze the following terms which do not fulfill the actual goal of citations and acknowledgements. i) Spurious citations ii) Biased citations iii) Self citations iv) Coercive citations. (08 Marks)
- b. What are the things author should acknowledge? (06 Marks)
- c. Elaborate on the following : (06 Marks)
 - i) Technical reading
 - ii) Critical and creative reading.

Module-3

- 5 a. What are Intellectual Property Rights? Explain the necessity of it. (06 Marks)
- b. Differentiate between invention and Innovation. What is a Patent? What are the criteria of patentability? (08 Marks)
- c. Enumerate the procedure for application preparation filing and grant of patents. (06 Marks)

OR

- 6 a. List and explain in detail about various types of Intellectual property rights. (08 Marks)
- b. Explain in detail about the infringement of patents. (06 Marks)
- c. What types of invention are not patentable in India? (06 Marks)

Module-4

- 7 a. What is Copyright? Explain classes, criteria and ownership of copyright. (08 Marks)
 b. Discuss about the copyright ownership issues. (06 Marks)
 c. Describe the process involved in the registration of a trademark. (06 Marks)

OR

- 8 a. Enumerate the procedure for registration of copyright. (08 Marks)
 b. Explain about the Trademark and rights from trademarks registration. (06 Marks)
 c. Explain about types of Trademarks registered in India. (06 Marks)

Module-5

- 9 a. How can industrial designs be protected? (06 Marks)
 b. What is a geographical indication? Explain the following with respect to GI.
 i) Ownership and right granted to the GI holders
 ii) Registered GI in India (08 Marks)
 c. Explain the following :
 i) Famous Industrial Designs
 ii) Generic GI and Homonymous GI (06 Marks)

OR

- 10 a. Explain the classification of Industrial Designs. (06 Marks)
 b. Explain the procedure for GI registration. (08 Marks)
 c. Explain about the enforcement of IPR in India. (06 Marks)

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21CS61

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are seven broad categories of computer software? (10 Marks)
b. What are the elements of a software process, generic process framework? (10 Marks)

OR

- 2 a. List the types of prescriptive process models and explain waterfall model and incremental process models. (10 Marks)
b. Explain spiral model and concurrent models. (10 Marks)

Module-2

- 3 a. Define requirements engineering and explain seven different tasks of requirements engineering. (10 Marks)
b. Draw and explain UMC use case diagram for SafeHome home security function. (10 Marks)

OR

- 4 a. Identify different ways of requirements model. Explain scenario based. (10 Marks)
b. Draw and explain activity diagram for access camera surveillance via the internet display camera views function. (10 Marks)

Module-3

- 5 a. Explain 12 agility principles to achieve agility. (10 Marks)
b. Explain SCRUM and FDD. (10 Marks)

OR

- 6 a. Explain Core Principles. (10 Marks)
b. Explain Construction principles. (10 Marks)

Module-4

- 7 a. Explain activities covered in software project management. (10 Marks)
b. Explain ways of categorizing software projects. (10 Marks)

OR

- 8 a. Explain Project charter. (10 Marks)
b. Explain setting objectives. (10 Marks)

Module-5

- 9 a. Explain place of software quality in project planning. (10 Marks)
b. Explain Boehm's model. (10 Marks)

OR

- 10 a. Explain V – Process Model. (10 Marks)
b. Explain software Reliability. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, 42+8 = 50, will be treated as malpractice.

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21CS62

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Full Stack Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the processing of Django request and response with a flow diagram. (10 Marks)
- b. Write a python function in Django framework that displays date and time which is an offset of n hours from current date and time. (10 Marks)

OR

- 2 a. Illustrate the process of URL mapping with views. (10 Marks)
- b. Describe MVC design pattern with a neat diagram. (10 Marks)

Module-2

- 3 a. Define the following: (i) Tags (ii) Filters (10 Marks)
- b. Describe the process of template system usage in Django. (10 Marks)

OR

- 4 a. Explain the following:
(i) Selecting objects
(ii) Deleting objects (10 Marks)
- b. Summarize the concept of template loading in Django. (10 Marks)

Module-3

- 5 a. Illustrate the usage of admin interface in Django. (10 Marks)
- b. Develop a Model form for student that contains his topic chosen for project, languages used and duration with a model called project. (10 Marks)

OR

- 6 a. Describe the following:
(i) Creating a feedback form
(ii) Processing the submission (10 Marks)
- b. Create a Django app that performs student registration course. Register admin interfaces, perform migrations and illustrate data entry through admin forms. (10 Marks)

Module-4

- 7 a. Develop a Django app that performs CSV and pdf generation for any models created. (10 Marks)
- b. Illustrate the generic views of objects. (10 Marks)

OR

- 8 a. Create a generic class view which displays list of students and detail view that displays student details for any selected student in the list. (10 Marks)
- b. Explain the syndication feed framework in generating Non-HTML content. (10 Marks)

Module-5

- 9 a. Define the following: (i) HTML (ii) XML (iii) JSON (iv) CSS (10 Marks)
b. Mention the jQuery Ajax facilities and explain any four. (10 Marks)

OR

- 10 a. Illustrate the using of jQuery UI Autocomplete in Django templates. (10 Marks)
b. Explain XML HTTP Request by mentioning the properties. (10 Marks)

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21CS63

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Computer Graphics and Fundamentals of Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain any five of the applications of computer graphics. (06 Marks)
- b. Develop the Bresenham's line drawing algorithm. Apply this algorithm for the line end points (20, 10) and (30, 18), list the points and plot the resultant line in x-y plane. (09 Marks)
- c. Explain OpenGL point and line functions with examples. (05 Marks)

OR

- 2 a. Explain the raster scan with neat diagram. Explain the role of the graphic controller. (06 Marks)
- b. Explain OpenGL functions to draw point, line and polygon objects in 2D. Develop the code snippet for drawing object star shape in 2D. (09 Marks)
- c. Illustrate the display window management using GLUT with diagram. (05 Marks)

Module-2

- 3 a. Explain the equation and matrix representation for a 2D object translation and rotation about the origin with the neat diagram. (08 Marks)
- b. With neat diagram, explain the 3D translation of a point and extend the same for 3D object. Write the equations for each vertex and matrix representation. (10 Marks)
- c. Justify the need of homogeneous coordinates. (02 Marks)

OR

- 4 a. Explain with the diagram the 3D object rotations and scaling. Give the matrix representations of the object. (08 Marks)
- b. Illustrate with neat diagram, the five steps sequence of transformations for rotating an 3D object about an axis parallel to the x-axis. (10 Marks)
- c. Define 2D reflection and shear. (02 Marks)

Module-3

- 5 a. Explain the LOCATOR, STROKE, VALUATOR, CHOICE and STRING Logical Input Devices. (08 Marks)
- b. Explain the basic approach to the design of animation sequences. (07 Marks)
- c. Explain the GLUT mouse functions with a program snippet. (05 Marks)

OR

- 6 a. Explain any three interactive picture-construction techniques with necessary diagram. (08 Marks)
- b. Explain the traditional animation techniques with bouncing-ball illustration. (07 Marks)
- c. Explain the GLUT keyboard functions with a program snippet. (05 Marks)

Module-4

- 7 a. Explain image processing and its related fields in detail. (10 Marks)
 b. Define pixel, resolution and its bit-depth of an image. Explain the digital image representation in the computer system with neat diagram. (08 Marks)
 c. An image of size 2.5 inches by 2 inches is scanned at 150 dpi. Determine the number of pixels in the image. (02 Marks)

OR

- 8 a. Explain the fundamental steps in image processing with diagram. (07 Marks)
 b. Explain the following terms or concepts with suitable examples with respect to pixels in an image:
 (i) Neighbourhood (ii) Connectivity (iii) Relations (11 Marks)
 c. Consider a 375×300 grey-scale image needs to be sent across the channel of capacity 28 Kbps, then determine the transmission time required. (02 Marks)

Module-5

- 9 a. Define image segmentation. Explain the classification of the segmentation algorithms. (10 Marks)
 b. Explain the following grey level discontinuities in a digital image:
 (i) Point detection (ii) Line detection (10 Marks)

OR

- 10 a. Explain edge detection and different stages in edge detection process. (10 Marks)
 b. Explain the following:
 (i) Sobel operator (ii) Canny edge detection (10 Marks)

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Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Advanced Java Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are enumerations? How to use enum constructor, instance variable and method? Explain with an example. (08 Marks)
- b. Explain how to obtain all annotations associated with a class and a method with a example. (08 Marks)
- c. Explain the various typewrappers used in java. (04 Marks)

OR

- 2 a. Explain the following with respect to annotations :
 - (i) Single member Annotations (08 Marks)
 - (ii) Default values in annotations. (06 Marks)
- b. What is autoboxing? Write a java program that demonstrates autoboxing and unboxing occurs inside expressions. (06 Marks)
- c. What are Annotations? Explain the following built-in annotations with examples :
 - (i) @Override (06 Marks)
 - (ii) @Target
 - (iii) @SuppressWarnings

Module-2

- 3 a. What are generics? Why Generics are required? Develop a simple program to show the concept of generics. (08 Marks)
- b. What are Bounded types? Demonstrate the use of bounded types with example program. (06 Marks)
- c. Describe Overriding methods in a generic class. (06 Marks)

OR

- 4 a. Why wild cards are required in Generics? Explain with an example program. (08 Marks)
- b. What is type Erasure? Describe Bridge methods with code snippet. (06 Marks)
- c. Explain Generic Super Class and Generic Sub Class with code Snippet. (06 Marks)

Module-3

- 5 a. Explain the string comparison methods with suitable examples. (08 Marks)
- b. Explain various string constructors in java with example. (04 Marks)
- c. What is String Buffer? How it is differ from string? Explain any six buffer methods. (08 Marks)

OR

- 6 a. How Compare Toll method differs from Compare To Ignore Case () method? Develop a Java program to sort an array of string in descending order by ignoring case. (08 Marks)
- b. Explain different methods to modify the strings objects by using String Class with an example. (06 Marks)
- c. Describe different character extraction methods with example. (06 Marks)

Module-4

- 7 a. What is Servlet? Explain life cycle of Servlets. Differentiate between Servlets and JSP. (07 Marks)
- b. Develop a program using Servlets that contains HTML page to accept Username and address and Display Greeting message as "Hello Username-address, How are you" in the browser window. (07 Marks)
- c. What are Cookies? Explain how Cookies are handled in Servlets? Develop a program to create a Cookie with name "Username" and Cookie value "XYZ". And also, display stored cookie in a web page. (06 Marks)

OR

- 8 a. What is JSP? Explain different JSP tags with example. (07 Marks)
- b. What is Session Object? Develop a program to create and read session object using JSP. (07 Marks)
- c. List and explain core classes and interfaces that are provided in javax.servlet.http package. (06 Marks)

Module-5

- 9 a. Describe the various steps of JDBC process with code snippets. (08 Marks)
- b. What is JDBC? Explain the four different types of JDBC drivers. (06 Marks)
- c. What are Transactions? Develop a program to demonstrate how to process a database transaction. (06 Marks)

OR

- 10 a. Briefly explain statement object. Develop a java program to demonstrate execute Query () and executeUpdate () methods. (10 Marks)
- b. What is meta data? Develop a program to display table name, column, count and structure of the table with size and data types. (10 Marks)

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21CS71

Seventh Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025

Big Data Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the evolution of Big Data. (06 Marks)
- b. Explain the characteristics of Big Data. (04 Marks)
- c. Explain Data Architecture Design, with a neat diagram. (10 Marks)

OR

- 2 a. Explain Analytics Scalability to Big Data and Massive parallel processing platforms. (12 Marks)
- b. Explain Big Data Analytics applications with one case study. (08 Marks)

Module-2

- 3 a. List and explain the core components of Hadoop. (10 Marks)
- b. Explain Hadoop Distributed File System. (10 Marks)

OR

- 4 a. Define MapReduce Frame work and its functions. (06 Marks)
- b. Explain steps on the request to MapReduce and the types of process in MapReduce. (10 Marks)
- c. Explain in brief on Flume Hadoop Tool. (04 Marks)

Module-3

- 5 a. Explain about No SQL datastore and its characteristics. (10 Marks)
- b. Describe the principle of working of the CAP theorem. (10 Marks)

OR

- 6 a. Demonstrate the working of key-value store with an example. (10 Marks)
- b. Describe the features of MongoDB, and its industrial application. (10 Marks)

Module-4

- 7 a. Explain the process in MapReduce when client submitting a job, with a neat diagram. (10 Marks)
- b. Explain Hive Integration and workflow steps involved with a diagram. (10 Marks)

OR

- 8 a. Using HiveQL for the following :
i) Create a table with partition
ii) Add, rename and drop a partition to a table. (10 Marks)
b. What is PIG in BigData? Explain the feature of PIG. (10 Marks)

Module-5

- 9 a. Explain linear and non-linear relationship with essential graphs in machine learning. (10 Marks)
b. Write the block diagram of text mining process and explain its phases. (10 Marks)

OR

- 10 a. With a neat diagram, write the steps in K-means clustering. (10 Marks)
b. Explain the purpose of web usage analytics and the significance of web graphs. (10 Marks)

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21CS72

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Cloud Computing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, explain cloud computing and its historical development. (10 Marks)
- b. List the characteristics and benefits of cloud computing. (10 Marks)

OR

- 2 a. Explain in brief the services provided by the following cloud service provider:
 - i) Amazon web service
 - ii) Microsoft azure
 - iii) Google AppEngine. (10 Marks)
- b. Write a note on challenges in cloud computing. (10 Marks)

Module-2

- 3 a. Define virtualization and explain hardware level virtualization with its advantages. (10 Marks)
- b. Discuss the taxonomy of virtualization techniques at different levels. (10 Marks)

OR

- 4 a. What are the characteristics of virtualized environment? (10 Marks)
- b. Explain with a neat diagram Type-I and Type-II hypervisor. (10 Marks)

Module-3

- 5 a. Explain the different types of cloud. (10 Marks)
- b. What is IaaS? Explain its reference implementation with a neat diagram. (10 Marks)

OR

- 6 a. Explain the economics of the cloud. (10 Marks)
- b. What does the acronym SaaS mean? How does it relate to cloud computing? (10 Marks)

Module-4

- 7 a. Analyze the various cloud security risks that organization face when utilizing cloud computing services. (10 Marks)
- b. Explain the security risks posed by a management OS. (10 Marks)

OR

- 8 a. Discuss the traditional concept of trust and trust necessary for online activities. (10 Marks)
- b. Explain in detail virtual machine security. (10 Marks)

Module-5

- 9 a. Describe Amazon EC2 and its basic features. (10 Marks)
- b. Analyze how cloud computing technology can be applied to support remote ECG monitoring. (10 Marks)

OR

- 10 a. What is a bucket? What type of storage does it provide? (10 Marks)
- b. Examine the core components of AppEngine. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

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21CS734

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Block Chain Technology

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Block Chain. With the neat diagram explain the Network view of a block chain. (07 Marks)
- b. Explain the different types of block chain. (08 Marks)
- c. Mention the benefits and limitations of block chain. (05 Marks)

OR

- 2 a. Explain the methods that are used to achieve decentralization. (04 Marks)
- b. Describe the Decentralized organizations in block chain. (06 Marks)
- c. Define the following :
i) Decentralization ii) Centralized systems iii) Distributed systems
iv) Atomicity v) Cap theorem. (10 Marks)

Module-2

- 3 a. Discuss about the cryptographic hash functions and its properties. (10 Marks)
- b. Write a note on Merkle trees. (05 Marks)
- c. What is double-spending problem? How to overcome this problem in crypto currency? (05 Marks)

OR

- 4 a. Discuss about the distributed consensus protocol with an example. (07 Marks)
- b. Explain the incentive mechanisms used in Bitcoin. (06 Marks)
- c. Explain the three important properties of hash puzzles. (07 Marks)

Module-3

- 5 a. Illustrate the concept on Bitcoin transactions long with its syntax. (07 Marks)
- b. Explain how to execute a script in a stack-based programming language. (06 Marks)
- c. Summarize the applications of Bitcoin scripts. (07 Marks)

OR

- 6 a. Explain hot and cold storage for Bitcoin along with its solutions, in detail. (10 Marks)
- b. Discuss about the payment service for Bitcoin in detail. Also explain its steps involved in Bitcoin Mines. (10 Marks)

Module-4

- 7 a. Discuss the tasks involved in bitcoin miners. (10 Marks)
- b. Explain the mining incentives and strategies to be followed by miners. (10 Marks)

OR

- 8 a. Explain the functions of decentralized mixing in Bitcoin. (05 Marks)
b. Discuss in detail how to De-anonymize Bitcoin. (10 Marks)
c. Write a note on Zerocoin. (05 Marks)

Module-5

- 9 a. Define Smart Contract. Explain the properties of Ricardian contract Bowtie model with a neat diagram. (08 Marks)
b. Explain Ethereum block chain with a state transition function. (06 Marks)
c. Explain the types of accounts for ethereum. (06 Marks)

OR

- 10 a. Write a note on precompiled contracts. (08 Marks)
b. Explain EVM (Ethereum Virtual Machine) with a neat diagram. (08 Marks)
c. Mention the various functions of the Iterator functions. (04 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Internet of Things

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1. a. In the context of Evolution of IoT, with neat diagram, illustrate the sequence of technological developments leading to the shaping of the modern day IoT. (06 Marks)
 b. With diagram, explain enabling IoT and the complex Interdependence of technologies. (08 Marks)
 c. Differentiate between the following :
 (i) IoT versus M2M.
 (ii) IoT versus CPS.
 (iii) IoT versus WoT. (06 Marks)

OR

2. a. With respect to the IoT networking components, define the following :
 (i) IoT NODE.
 (ii) IoT Router.
 (iii) IoT LAN
 (iv) IoT Gateway
 (v) IoT Proxy (10 Marks)
 b. Discuss the following addressing strategies in IoT:
 (i) Address Management Classes. (10 Marks)
 (ii) Addressing during node Mobility. (10 Marks)

Module-2

3. a. Define sensors and with diagram, outline the simple sensing operation. (04 Marks)
 b. Discuss Scalar and Vector sensors and draw the functional blocks of a typical sensor node in IoT. (06 Marks)
 c. With neat diagram, explain the different sensing types commonly encountered in IoT. (10 Marks)

OR

4. a. Define Actuator and with diagram, discuss the outline of a simple actuation mechanism. (04 Marks)
 b. Explain the various actuators classes any 5 in IoT. (10 Marks)
 c. Discuss actuator characteristics that define all actuators. (06 Marks)

Module-3

5. a. List and discuss common data types used in IoT applications. (06 Marks)
 b. Explain the various processing topologies in IoT with necessary diagrams. (08 Marks)
 c. Illustrate the importance of processing in IoT. (06 Marks)

OR

- 6 a. With neat diagram, explain the processing offloading paradigm for the development of IoT-based solutions. (10 Marks)
- b. Determine the importance of choosing the right processing topologies and associated considerations while designing IoT applications. (10 Marks)

Module-4

- 7 a. List common connectivity protocols in IoT. (04 Marks)
- b. Explain the salient features and application scope of any 5 connectivity protocols. (10 Marks)
- c. Differentiate between Wi-Fi and Bluetooth connectivity protocols in IoT. (06 Marks)

OR

- 8 a. With necessary diagrams, explain in detail any four connectivity protocols in IoT. (10 Marks)
- b. Determine the requirements associated with any of IoT connectivity protocols in real-world solutions. (10 Marks)

Module-5

- 9 a. Describe in detail, the various Infrastructure protocols in IoT-based communication technologies. (10 Marks)
- b. Explain the following discovery protocols :
- (i) Physical Web
 - (ii) mDNS
 - (iii) Universal plug and play (UPnP). (10 Marks)

OR

- 10 a. With neat diagram, discuss the illustration of the various facets of the Interoperability in IoT. (10 Marks)
- b. Discuss any four IoT Interoperability standards. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 NOSQL Database

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the key difference between NOSQL and Relational databases. (10 Marks)
- b. Provide strategies for optimizing data models to improve application performance. (10 Marks)

OR

- 2 a. Explain impendence mismatch problem in the context of data storage, and how does it affect application development. (10 Marks)
- b. Describe the concept of schemaless databases and their benefits. (10 Marks)

Module-2

- 3 a. Explain the concepts of Master – Slave and Peer- to – peer replication in distributed databases. (10 Marks)
- b. Discuss the importance of version stamp in esuring consistency across multiple nodes in a distributed database. (10 Marks)

OR

- 4 a. Discuss the challenges associated with achieving update consistency in a distributed database system. (10 Marks)
- b. Discuss read consistency in distributed databases, considering factors such as staleness and isolation levels. (10 Marks)

Module-3

- 5 a. Provide an example of composing Map-Reduce calculations to process and Analyze data. (10 Marks)
- b. Discuss the scalability characteristics of key-value databases. (10 Marks)

OR

- 6 a. Discuss key features of key value stores and their advantages. (10 Marks)
- b. Describe the basic structure of data in a key value databases. (10 Marks)

Module-4

- 7 a. Explain fundamental principles of document database. (10 Marks)
- b. Discuss the importance of SEO (Search Engine Optimization) in the context of blogging platforms. (10 Marks)

OR

- 8 a. Provide examples of common query operations performed on document databases and explain their significance. (10 Marks)
- b. Explain the importance of event logging in web applications with examples. (10 Marks)

Module-5

- 9 a. Discuss the key features of graph databases that make them suitable for handling connected data. (10 Marks)
- b. Identify scenarios where using a graph database may not be appropriate what are the limitations. (10 Marks)

OR

- 10 a. Describe how transactions are handled in graph database. What are the ACID properties are implemented. (10 Marks)
- b. Provide examples of complex queries that can be efficiently executed in graph database. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Programming in Python

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain computer Hardware architecture with neat diagram. (06 Marks)
- b. Summarize the python terminology interpreter and compiler. (07 Marks)
- c. What is a program? Illustrate the building blocks of programs. (07 Marks)

OR

- 2 a. Develop a python program which prompts the user for a Celsius temperature, convert the temperature to fahrenheit and print out the converted temperature. (07 Marks)
- b. Demonstrate the python functions int(), float() and str() with example. (06 Marks)
- c. List and explain the significant features of python programming language. (07 Marks)

Module-2

- 3 a. Explain the different comparison and Boolean operators, along with examples. (08 Marks)
- b. Summarize syntax and example of using if-else ladder in python. (06 Marks)
- c. Develop a python program to check whether the given number is odd (or) even. (06 Marks)

OR

- 4 a. Experiment with iteration: while and for, give appropriate examples also for the same. (08 Marks)
- b. Explain the following with python programme:
i) break ii) Continue. (08 Marks)
- c. Write a program to find the factorial of a number. (04 Marks)

Module-3

- 5 a. Define function. Explain function calls in python programming with an example. (08 Marks)
- b. Explain the local and global scope of the variable with a suitable example. (06 Marks)
- c. Analyze following code and write the output of the following code with relevant explanation.

```
def fred():
    print("Zap")
def jane():
    print("ABC")
jane()
fred()
jane()
```

(06 Marks)

OR

- 6 a. Define String. Explain getting the length of a string using len. (06 Marks)
 b. Analyze the following with examples:
 i) String slicer
 ii) Strings are immutable. (10 Marks)
 c. Summarize string comparison with example. (04 Marks)

Module-4

- 7 a. Explain various list operations with an example. (08 Marks)
 b. Explain the following with examples:
 i) Lists and functions (08 Marks)
 ii) Lists and strings. (04 Marks)
 c. Explain tuples are immutable with an example.

OR

- 8 a. Explain Dictionaries operations and methods. (10 Marks)
 b. Explain the following:
 i) Tuple or return value
 ii) Tuple comprehension. (10 Marks)

Module-5

- 9 a. Explain the following with examples:
 i) Character matching in regular expressions. (12 Marks)
 ii) Extracting data using regular expressions. (08 Marks)
 b. Illustrate the Escape character.

OR

- 10 a. Explain the concept of file handling. Also explain reading and writing process with suitable example. (10 Marks)
 b. Explain the following:
 i) Errors and Exceptions
 ii) Handling Exceptions. (10 Marks)

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Mathematics – III for Computer Science Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Mathematics Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1									M	L	C		
Q.1	a.	A random variable x has the following prob. density function for various values of x .							07	L2	CO1		
		x	0	1	2	3	4	5				6	7
		P(x)	0	k	2k	2k	3k	k ²				2k ²	7k ² +k
Find the value of k and evaluate $P(x < 6)$, $P(3 < x \leq 6)$ and $(x \geq 6)$.													
	b.	Derive the mean and variance of Poisson distribution.							06	L2	CO2		
	c.	In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for? (i) less than 10 minutes (ii) more than 10 minutes and (iii) between 10 and 12 minutes.							07	L3	CO2		
OR													
Q.2	a.	The probability density function of $f(x) = \begin{cases} Kx^2, & -3 < x < 3 \\ 0, & \text{elsewhere} \end{cases}$ Find the value of K and evaluate (i) $P(x < 2)$, $P(x > 1)$ (ii) $P(1 \leq x \leq 2)$							07	L3	CO1		
	b.	When a coin is tossed 4 times, find the probability of getting (i) exactly one head (ii) atleast three heads and (iii) less than two heads.							06	L2	CO2		
	c.	The marks of 1000 students in an examination follows a normal distribution with mean > 0 and S.D 5. Find the number of students whose marks will be (i) less than 65 (ii) more than 75 and (iii) between 65 and 75.							07	L2	CO2		
Module – 2													
Q.3	a.	If the joint probability distribution of x and y is given by $f(x, y) = \frac{1}{30}(x + y), \text{ for } x = 0, 1, 2, 3; y = 0, 1, 2$ Find (i) $P(x \leq 2, y = 1)$ (ii) $P(x > y)$							07	L2	CO2		
	b.	Find the unique fixed probability vector of $P = \begin{bmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{bmatrix}$							06	L2	CO3		
	c.	Three boys A, B and C are throwing a ball to each other. A always throw the ball to B. B always throw the ball to A and C is just as likely to throw the ball to A as to B. Find the probability that C has the ball after three throws, if C starts the game.							07	L3	CO3		

OR

OR																						
Q.4	a.	The joint prob. distribution for the following data, find E(x) and E(y). <table border="1"><tr><td>Y \ X</td><td>-2</td><td>-1</td><td>4</td><td>5</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.0</td><td>0.3</td></tr><tr><td>2</td><td>0.2</td><td>0.1</td><td>0.1</td><td>0</td></tr></table>			Y \ X	-2	-1	4	5	1	0.1	0.2	0.0	0.3	2	0.2	0.1	0.1	0	07	L2	CO2
Y \ X	-2	-1	4	5																		
1	0.1	0.2	0.0	0.3																		
2	0.2	0.1	0.1	0																		
	b.	Show that the matrix $P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix.			06	L2	CO3															
	c.	A gambler's luck follows pattern. If he wins a game the prob. of winning the next game is 0.6. However, if he loses a game, the prob. of losing the next game is 0.7. There is an even chance of the gambler winning the first game. What is the prob. of he winning the second game.			07	L3	CO3															
Module – 3																						
Q.5	a.	Define (i) Null hypothesis (ii) A statistic (iii) Standard error (iv) Level of significance (v) Test of significance.			07	L1	CO4															
	b.	A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% LOS.			06	L3	CO4															
	c.	In a city A 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 5% significance level?			07	L3	CO5															
OR																						
Q.6	a.	Explain the following terms: (i) Type-I and Type-II errors (ii) Statistical hypothesis (iii) Critical region (iv) Alternate hypothesis			07	L1	CO4															
	b.	The average marks in Engg. Maths of a sample of 100 students was 51 with S.D 6 marks. Could this have been a random sample from a population with average marks 50?			06	L2	CO5															
	c.	One type of aircraft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of aircrafts so far as engine defects are concerned? Test at 0.05 significance level.			07	L3	CO4															
Module – 4																						
Q.7	a.	State central limit theorem. Use the theorem to evaluate $P(50 < x < 56)$ where $\bar{x}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$.			07	L2	CO4															
	b.	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find 95% confidence interval for the population mean. Given that $Z(0.15) = 0.0596$.			06	L2	CO5															
	c.	Fit a Poisson distribution to the following data and test for goodness of fit at 5% LOS. <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f</td><td>419</td><td>352</td><td>154</td><td>56</td><td>19</td></tr></table>			x	0	1	2	3	4	f	419	352	154	56	19	07	L3	CO5			
x	0	1	2	3	4																	
f	419	352	154	56	19																	

OR

Q.8	a.	Height of a random sample of 50 college student showed a mean of 174.5 cms and a S.D 6.9 cms. Construct 99% confidence limits for the mean height of all college students.	07	L2	CO4
	b.	A random sample of 10 boys had the following I.Q : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. DO these data support the assumption of a population mean I.Q of 100 (at 5% LOS)?	06	L3	CO5
	c.	The theory predicts the propositions of beans in the four groups, G_1 , G_2 , G_3 , G_4 should be in the ratio 9 : 3 : 3 : 1. In experiment with 1600 beans the numbers in the groups were 882, 313, 287 and 118. Does the experimental support the theory.	07	L3	CO5

Module – 5

Q.9	a.	The varieties of wheat A, B, C were shown in four plots each and the following yields in quintals per acre were obtained. <table><tr><td>A</td><td>8</td><td>4</td><td>6</td><td>7</td></tr><tr><td>B</td><td>7</td><td>6</td><td>5</td><td>3</td></tr><tr><td>C</td><td>2</td><td>5</td><td>4</td><td>4</td></tr></table> Test the significance of difference between the yields of varieties, given that 5% tabulated value of $F = 4.26$ with (2, 9) d.f. Set up one-way ANOVA and using direct method.	A	8	4	6	7	B	7	6	5	3	C	2	5	4	4	10	L3	CO6										
A	8	4	6	7																										
B	7	6	5	3																										
C	2	5	4	4																										
	b.	Present your conclusion after doing ANOVA to the following results of the Latin-square design conducted in respect of five fertilizers which were used on plots of different fertility. <table><tr><td>A(16)</td><td>B(10)</td><td>C(11)</td><td>D(9)</td><td>E(9)</td></tr><tr><td>E(10)</td><td>C(9)</td><td>A(14)</td><td>B(12)</td><td>D(11)</td></tr><tr><td>B(15)</td><td>D(8)</td><td>E(8)</td><td>C(10)</td><td>A(18)</td></tr><tr><td>D(12)</td><td>E(6)</td><td>B(13)</td><td>A(13)</td><td>C(12)</td></tr><tr><td>C(13)</td><td>A(11)</td><td>D(10)</td><td>E(7)</td><td>B(14)</td></tr></table>	A(16)	B(10)	C(11)	D(9)	E(9)	E(10)	C(9)	A(14)	B(12)	D(11)	B(15)	D(8)	E(8)	C(10)	A(18)	D(12)	E(6)	B(13)	A(13)	C(12)	C(13)	A(11)	D(10)	E(7)	B(14)	10	L3	CO6
A(16)	B(10)	C(11)	D(9)	E(9)																										
E(10)	C(9)	A(14)	B(12)	D(11)																										
B(15)	D(8)	E(8)	C(10)	A(18)																										
D(12)	E(6)	B(13)	A(13)	C(12)																										
C(13)	A(11)	D(10)	E(7)	B(14)																										

OR

Q.10	a.	Set up two-way ANOVA table for the data given below, using coding method subtracting 40 from the given numbers. <table><tr><th rowspan="2">Pieces of land</th><th colspan="4">Treatment</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P</td><td>45</td><td>40</td><td>38</td><td>37</td></tr><tr><td>Q</td><td>43</td><td>41</td><td>45</td><td>38</td></tr><tr><td>R</td><td>39</td><td>39</td><td>41</td><td>41</td></tr></table>	Pieces of land	Treatment				A	B	C	D	P	45	40	38	37	Q	43	41	45	38	R	39	39	41	41	10	L3	CO6
Pieces of land	Treatment																												
	A	B	C	D																									
P	45	40	38	37																									
Q	43	41	45	38																									
R	39	39	41	41																									
	b.	There are three main brands of a certain power. A set of its 120 sales is examined and found to be allocated among four groups (A, B, C, D) and brands (I, II, III) as follows: <table><tr><th rowspan="2">Brands</th><th colspan="4">Groups</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>I</td><td>0</td><td>4</td><td>8</td><td>15</td></tr><tr><td>II</td><td>5</td><td>8</td><td>13</td><td>6</td></tr><tr><td>III</td><td>18</td><td>19</td><td>11</td><td>13</td></tr></table> Is there any significant difference in brands preference? Answer at 5% level, using one-way ANOVA. Take 10 as the code value to subtract it from all given values.	Brands	Groups				A	B	C	D	I	0	4	8	15	II	5	8	13	6	III	18	19	11	13	10	L3	CO6
Brands	Groups																												
	A	B	C	D																									
I	0	4	8	15																									
II	5	8	13	6																									
III	18	19	11	13																									

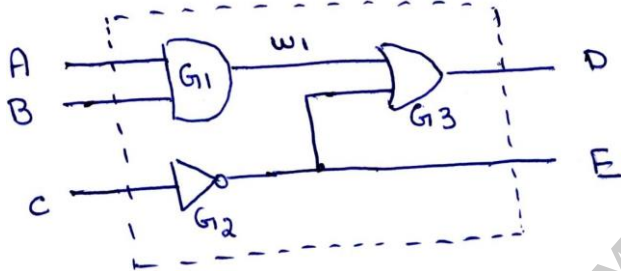
Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Digital Design and Computer Organization

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Determine the complement of the following function: (i) $F = xy' + x'y$ (ii) $F = x'yz' + x'y'z$		06	L3	CO1
	b.	Describe map method for three variables.		04	L2	CO1
	c.	Apply K map technique to simplify the following function: (i) $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$ (ii) $F(x, y, z) = x'y + yz' + y'z'$		10	L3	CO1
OR						
Q.2	a.	Apply K map technique to simplify the function : $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ and $d(w, x, y, z) = \Sigma(0, 2, 5)$		06	L3	CO1
	b.	Determine all the prime implicants for the Boolean function F and also determine which are essential $F(w, x, y, z) = \Sigma(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$		10	L3	CO1
	c.	Develop a verilog gate-level description of the circuit shown in Fig.Q2(c). 		04	L3	CO1
Module – 2						
Q.3	a.	Explain the combinational circuit design procedure with code conversion example.		10	L2	CO2
	b.	Design a full adder circuit. Also develop data flow verilog model for full adder.		10	L3	CO2
OR						
Q.4	a.	Describe 4×1 MUX with block diagram and truth table. Also develop a behavioral model verilog code for 4×1 MUX.		10	L2	CO2
	b.	What are storage elements? Explain the working of SR and D latch along with logic diagram and function table.		10	L2	CO2
Module – 3						
Q.5	a.	Explain the basic operational concepts between the processor and memory.		10	L2	CO3
	b.	Describe the following: (i) Processor clock (ii) Basic performance equation (iii) Clock rate (iv) SPEC rating		10	L2	CO3
OR						
Q.6	a.	Define addressing mode. Explain any four types of addressing mode with example.		10	L2	CO3

	b.	Mention four types of operations to be performed by instructions in a computer. Explain the basic types of instruction formats to carry out. $C \leftarrow [A] + [B]$	10	L2	CO3
Module – 4					
Q.7	a.	With a neat diagram, explain the concept of accessing I/O devices.	10	L2	CO4
	b.	What is bus arbitration? Explain centralized and distributed arbitration method with a neat diagram.	10	L2	CO4
OR					
Q.8	a.	With neat sketches, explain various methods for handling multiple interrupts requests raised by multiple devices.	10	L2	CO4
	b.	What is cache memory? Explain any two mapping function of cache memory.	10	L2	CO4
Module – 5					
Q.9	a.	Draw the single bus architecture and write the control sequence for execution of instruction ADD (R ₃), R ₁ .	10	L3	CO5
	b.	With suitable diagram, explain the concept of register transfer and fetching of word from memory.	10	L2	CO5
OR					
Q.10	a.	With a neat diagram, explain the flow of 4-stage pipeline operation.	10	L2	CO5
	b.	Explain the role of cache memory and pipeline performance.	10	L2	CO5

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Operating Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C												
Q.1	a.	Define Operating System. Explain dual mode of operating systems with a neat diagram.	06	L1 L2	CO1												
	b.	Distinguish between the following terms: i) Multiprogramming and Multitasking ii) Multiprocessor and Clustered system	06	L2	CO1												
	c.	Explain with a neat diagram VM-WARE Architecture.	08	L1 L2	CO1												
OR																	
Q.2	a.	List and explain the services provided by OS for the user and efficient operation of system.	06	L2	CO1												
	b.	Explain the different computing equipments.	06	L2	CO1												
	c.	What are systems calls? List and explain the different types of systems calls.	08	L1 L2	CO1												
Module – 2																	
Q.3	a.	What is process? Explain process state diagram and process control block with a neat diagram.	10	L1 L2	CO2												
	b.	What is interprocess communication? Explain direct and indirect communication with respect to message passing system.	10	L1 L2	CO2												
OR																	
Q.4	a.	List and explain the different types of multithreading models.	06	L1 L2	CO2												
	b.	Calculate the average waiting time and average turnaround time by drawing the Gantt-chart using FCFS, SJF, RR (Q = 4ms) and priority scheduling (Higher Number is having highest priority). <table border="1"><tr><td>Process</td><td>B.T. (ms)</td><td>Priority</td></tr><tr><td>P₁</td><td>24</td><td>1</td></tr><tr><td>P₂</td><td>03</td><td>2</td></tr><tr><td>P₃</td><td>03</td><td>3</td></tr></table>	Process	B.T. (ms)	Priority	P ₁	24	1	P ₂	03	2	P ₃	03	3	14	L3	CO2
Process	B.T. (ms)	Priority															
P ₁	24	1															
P ₂	03	2															
P ₃	03	3															
Module – 3																	
Q.5	a.	What is critical section? Give the Peterson’s solution to 2 processes critical section problem.	05	L1 L2	CO3												
	b.	Explain Reader’s and Writer’s problem in detail.	07	L2	CO3												
	c.	What is semaphore? Discuss the solution to the classical dinning philosopher problem.	08	L1 L2	CO3												

OR																																																																										
Q.6	a.	What is a Deadlock? What are the necessary conditions for the deadlock to occur?	06	L1 L2	CO3																																																																					
	b.	Consider the following snap shot of the system. <table><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th colspan="3">Max</th><th colspan="3">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th><th>A</th><th>B</th><th>C</th></tr><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> Answer the following questions: i) What is the content of the matrix need? ii) Is the system on a safe state? If so, find safe sequence. iii) If P ₁ requirements for (1, 0, 2) additional resources can P ₁ be granted.	Process	Allocation			Max			Available			A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3				14	L3	CO2
Process	Allocation			Max			Available																																																																			
	A	B	C	A	B	C	A	B	C																																																																	
P ₀	0	1	0	7	5	3	3	3	2																																																																	
P ₁	2	0	0	3	2	2																																																																				
P ₂	3	0	2	9	0	2																																																																				
P ₃	2	1	1	2	2	2																																																																				
P ₄	0	0	2	4	3	3																																																																				
Module – 4																																																																										
Q.7	a.	What is paging? Explain with a neat diagram paging hardware with TLB.	10	L1 L2	CO4																																																																					
	b.	Explain the different strategies used to select a free hole from available holes.	05	L1	CO4																																																																					
	c.	What is Fragmentation? List and explain its types.	05	L2	CO4																																																																					
OR																																																																										
Q.8	a.	What is page fault? With a neat diagram explain steps in handling page fault.	08	L2	CO4																																																																					
	b.	Consider the page reference string for a memory with 3 frames determine the number of page faults using FIFO, optimal and LRU replacement algorithms. Which algorithms is more efficient? 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1	12	L3	CO4																																																																					
Module – 5																																																																										
Q.9	a.	Define File. List and explain different file operations and file attributes.	10	L1	CO5																																																																					
	b.	Explain the different file allocation methods.	10	L2	CO5																																																																					
OR																																																																										
Q.10	a.	What is Access Matrix? Explain the implementation of Access Matrix.	10	L2	CO5																																																																					
	b.	A drive has 5000 cylinders numbered 0 to 4999. The drive is currently servicing at a request 143 and previously served a request at 125. The queue of pending request in FIFO order. 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 starting from current head position. What is the total distance travelled (in cylinders) by a disk arm to satisfy the request using FCFS, SSTF, SCAN, LOOK and C-Look algorithm	10	L3	CO5																																																																					

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Define Data Structures. Explain the classification of data structures with a neat diagram.	8	L2	CO1	
	b.	Write a C Functions to implement pop , push and display operations for stacks using arrays.	7	L2	CO2	
	c.	Differentiate structures and unions.	5	L2	CO1	
OR						
Q.2	a.	Write an algorithm to evaluate a postfix expression and apply the same for the given postfix expression. 6 2 / 3 – 4 2 * +.	7	L3	CO2	
	b.	Explain the dynamic memory allocation function in detail.	8	L2	CO1	
	c.	What is Sparse matrix? Give the triplet form of a given matrix and find its transpose $A = \begin{bmatrix} 0 & 0 & 3 & 0 & 4 \\ 0 & 0 & 5 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 6 & 0 & 0 \end{bmatrix}$	5	L3	CO1	
Module – 2						
Q.3	a.	Define Queue. Discuss how to represent a queue using dynamic arrays.	8	L2	CO2	
	b.	Write a C Function to implement insertion () , deletion () and display () operations on circular queue.	6	L3	CO2	
	c.	Write a note on Multiple stacks and queues with suitable diagram.	6	L2	CO2	
OR						
Q.4	a.	What is a linked list? Explain the different types of linked list with neat diagram.	6	L2	CO3	
	b.	Write a C function for the following on singly linked list with example : i) Insert a node of the beginning ii) Delete a node at the front iii) Display.	8	L3	CO3	
	c.	Write the C function to add two polynomials.	6	L2	CO3	

Module – 3					
Q.5	a.	Discuss how binary trees are represented using : i) Assay ii) Linked list.	6	L2	CO4
	b.	Define Threaded binary tree. Discuss In – threaded binary tree.	6	L2	CO4
	c.	Write the C function for the following additional list operation : i) Inverting Singly linked list ii) Concatenating Singly linked list.	8	L3	CO3
OR					
Q.6	a.	Discuss Inorder , Preorder , Postorder and Level order traversal with suitable function for each.	8	L3	CO4
	b.	Define the threaded binary tree. Construct threaded binary tree for the following element : A, B, C, D, E, F, G, H, I.	6	L2	CO4
	c.	Write a C function for the following : i) Insert a node at the beginning of doubly linked list. ii) Deleting a node at the end of the doubly linked list.	6	L3	CO3
Module – 4					
Q.7	a.	Define Forest , Transform the forest into a binary tree and traverse using inorder , preorder and postorder traversal with an example.	8	L1	CO5
	b.	Define Binary search tree. Construct a binary search tree for the following elements : 100 , 85 , 45 , 55 , 120 , 20 , 70 , 90 , 115 , 65 , 130 , 145.	6	L2	CO5
	c.	Discuss Selection tree with an example.	6	L2	CO5
OR					
Q.8	a.	Define Graph. Explain adjacency matrix and adjacency list representation with an example.	8	L2	CO5
	b.	Define the following terminology with example : i) Digraph ii) Weighted graph iii) Self loop iv) Connected graph.	6	L2	CO5
	c.	Briefly explain about Elementary graph operations.	6	L3	CO5
Module – 5					
Q.9	a.	Explain in detail about Static and Dynamic Hashing.	6	L2	CO5
	b.	What is Collision? What are the methods to resolve collision?	7	L2	CO5
	c.	Explain Priority queue with the help of an examples.	7	L2	CO5
OR					
Q.10	a.	Define Hashing. Explain different hashing functions with suitable examples.	12	L2	CO5
	b.	Write short note on : i) Leftist trees ii) Optimal binary search tree.	8	L3	CO5

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Object Oriented Programming with JAVA

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	List and explain any three features of object oriented programming.	6	L1	CO1	
	b.	What do you mean by type conversion and type casting? Give examples.	8	L2	CO1	
	c.	How to declare and initialize 1-D and 2-D arrays in Java. Give examples.	6	L2	CO1	
OR						
Q.2	a.	List the short circuit operators and show the concept using few examples.	4	L2	CO1	
	b.	With a java program, illustrate the use of ternary operator to find the greatest of three numbers.	6	L3	CO1	
	c.	Develop a Java program to demonstrate the working of for each version of for loop. Initialize the 2D array with values and print them using for each.	10	L2	CO1	
Module – 2						
Q.3	a.	Develop a program in Java to implement a stack of integers.	12	L3	CO2	
	b.	What are constructors? Give the types and explain the properties of constructors. Support with appropriate examples.	8	L2	CO2	
OR						
Q.4	a.	Illustrate with an example program to pass objects as arguments.	10	L2	CO2	
	b.	Explain different access specifiers in Java with example program.	10	L2	CO2	
Module – 3						
Q.5	a.	Define inheritance. List and explain different types of inheritance in Java with code snippets.	10	L2	CO3	
	b.	Compare and contrast between overloading and overriding in Java with example program for each.	10	L2	CO3	
OR						
Q.6	a.	Analyze an interface in Java and list out the speed of an interface. Illustrate with the help of a program the importance of an interface.	10	L2	CO3	
	b.	List the different uses of final and demonstrate each with the of code snippets.	10	L2	CO3	

1 of 2

Module – 4

Q.7	a.	Define a package. Explain how to create user defined package with example.	7	L2	CO4
	b.	Discuss about exception handling in Java. Give the framework of the exception handling block. List the types of exception.	8	L2	CO4
	c.	Develop a Java program to raise a custom exception for division by zero using try, catch, throw and finally.	5	L3	CO4

OR

Q.8	a.	Compare throw and throws keyword by providing suitable example program.	10	L2	CO4
	b.	Explain about the need for finally block.	5	L2	CO4
	c.	Discuss about chained exceptions.	5	L2	CO4

Module – 5

Q.9	a.	Define thread. Demonstrate creation of multiple threads with a program.	10	L2	CO5
	b.	Explain the two ways in which Java threads can be instantiated. Support your explanation with a sample program.	10	L2	CO5

OR

Q.10	a.	What is enumeration? Explain the methods values() and valueof().	10	L2	CO5
	b.	Explain about type wrappers and auto boxing.	10	L2	CO5

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Object Oriented Programming with C++

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is meant by constructor? Discuss the types of constructors in C++ with example.	07	L2	CO1
	b.	Demonstrate with example, how friend functions, friend classes and inline functions are useful in C++.	07	L2	CO1
	c.	With a sample program, demonstrate the general format of a C++ program.	06	L3	CO1
OR					
Q.2	a.	List the various access specifiers supported by OOPs. Illustrate their use.	07	L2	CO1
	b.	What is the significance of static data and member functions in C++? Explain.	07	L2	CO1
	c.	What is Object Oriented Programming? Explain its features.	06	L2	CO1
Module – 2					
Q.3	a.	Develop a object oriented program to find the smallest and biggest among array elements.	07	L3	CO2
	b.	Discuss pointers to object with example code. Also discuss their advantages.	07	L2	CO2
	c.	Explain different array type in C++ with suitable example code snippet.	06	L3	CO2
OR					
Q.4	a.	What is “this” pointer? Illustrate the use of “this” pointer in C++.	07	L2	CO2
	b.	Discuss when two or more functions are said to be overloaded. Identify the causes of ambiguity in function overloading.	07	L2	CO2
	c.	What is a dynamic constructor? Explain with an example program.	06	L3	CO2
Module – 3					
Q.5	a.	List the operators in C++ that can not be overloaded. Develop a C++ program using “Time” class, to overload the ‘+’ and ‘-’ operators.	07	L3	CO3
	b.	Illustrate the role of access-specifiers in different level of inheritances.	07	L2	CO3
	c.	Discuss various types of inheritances with suitable example codes.	06	L3	CO3
OR					
Q.6	a.	Illustrate the use of constructors and destructors in inheritance in C++.	07	L2	CO3
	b.	Develop a program in C++ to derive a class “Manager” from class “Person” and “Employee”. Consider suitable data members and member-functions for the classes.	07	L3	CO3
	c.	Explain “Virtual base Class” with an example.	06	L3	CO2

Module – 4					
Q.7	a.	Design a C++ program demonstrating the use of the Pure Virtual function using base and derived classes. Also explain the code.	07	L3	CO4
	b.	What is polymorphism in C++? Explain its types with example.	07	L3	CO4
	c.	Explain virtual function in C++. Discuss what is early and late binding.	06	L2	CO4
OR					
Q.8	a.	What are generic functions? Demonstrate the use of generic function in swapping two variables of any type.	07	L2	CO4
	b.	What is typename and export keyword? Discuss their usage. Discuss the advantages of using templates in C++.	07	L2	CO4
	c.	Explain the use of a class template. Also explain class template with suitable code snippet.	06	L3	CO4
Module – 5					
Q.9	a.	Explain the fundamentals of exception handling in C++. Analyze the benefits of exception handling.	07	L2	CO5
	b.	Discuss different standard exceptions in C++.	07	L2	CO5
	c.	Develop a C++ program to demonstrate the usage of try, catch and through to handle exceptions.	06	L3	CO5
OR					
Q.10	a.	What are the different file opening modes in C++? Compare and contrast file opening modes.	07	L2	CO5
	b.	Explain file streams with example.	06	L3	CO5
	c.	Develop a C++ program to create a text file, check file is created or not, if created, write some text in to the file and the read and display the text from the file.	07	L3	CO5

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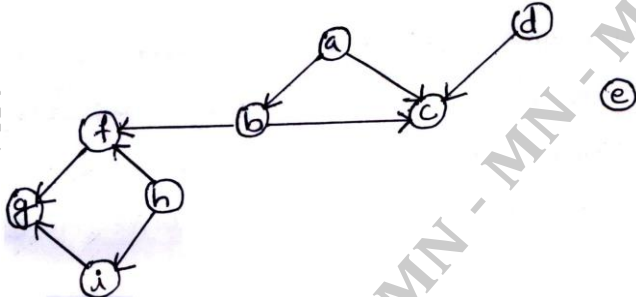
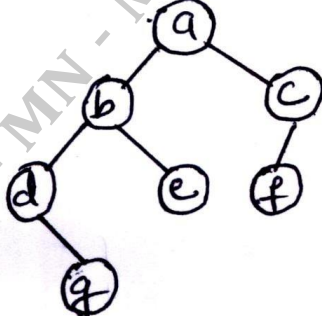
Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Analysis and Design of Algorithms

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the various steps in algorithm design and analysis process with the flow diagram.	08	L1	CO1
	b.	Give formal and informal definitions of asymptotic notations.	06	L1	CO1
	c.	Explain the general plan of mathematical analysis of recursive algorithm with an example.	06	L1	CO1
OR					
Q.2	a.	Design algorithm for tower of Hanoi problem and obtain time complexity.	10	L1	CO1
	b.	Write an algorithm to search an element in an array using sequential search. Discuss the best case, worst case and average case efficiency of this algorithm.	10	L1	CO1
Module – 2					
Q.3	a.	Write an algorithm to sort the numbers using insertion sort. Discuss its efficiency.	10	L2	CO2
	b.	Design quick sort algorithm and obtain its best, average and worst case efficiency.	10	L2	CO2
OR					
Q.4	a.	Write merge sort algorithm and sort the list E X A M P L E.	08	L2	CO2
	b.	Apply the DFS based algorithm to solve the topological sorting problem for the following graph, Fig.Q4(b)	06	L3	CO2
					
Fig.Q4(b)					
	c.	Write algorithm for pre-order, post order and in order traversals of a tree. Write pre-order, in-order and post order for the given tree.	06	L2	CO2
					
Fig.Q4(c)					

1 of 3

Module – 3

Q.5	a.	Define AVL tree. Construct AVL tree for the list 5, 6, 8, 3, 2, 4, 7.	10	L3	CO3
	b.	Define heap. Sort the following lists by heapsort: H E A P S O R T (in alphabetical order)	10	L3	CO3

OR

Q.6	a.	Write the algorithm for comparison counting sort. Discuss its efficiency.	10	L2	CO4
	b.	Design Horspools algorithm for string matching. Apply Horspools algorithm to find the pattern BARBER on the text JIM_SAW_ME_IN_BARBERSHOP	10	L3	CO4

Module – 4

Q.7	a.	Write Warshall's algorithm and apply the same to compute transitive closure of a directed graph. <table border="1"> <tr> <td></td> <td>a</td> <td>b</td> <td>c</td> <td>d</td> <td>e</td> </tr> <tr> <td>a</td> <td>1</td> <td>0</td> <td>0</td> <td>1</td> <td>0</td> </tr> <tr> <td>b</td> <td>0</td> <td>1</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>c</td> <td>0</td> <td>0</td> <td>0</td> <td>1</td> <td>1</td> </tr> <tr> <td>d</td> <td>1</td> <td>0</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>e</td> <td>0</td> <td>1</td> <td>0</td> <td>0</td> <td>1</td> </tr> </table>		a	b	c	d	e	a	1	0	0	1	0	b	0	1	0	0	0	c	0	0	0	1	1	d	1	0	0	0	0	e	0	1	0	0	1	10	L3	CO3
	a	b	c	d	e																																				
a	1	0	0	1	0																																				
b	0	1	0	0	0																																				
c	0	0	0	1	1																																				
d	1	0	0	0	0																																				
e	0	1	0	0	1																																				

- b. Construct minimum cost spanning tree using Kruskal's algorithm for the following graph, Fig.Q7(b).

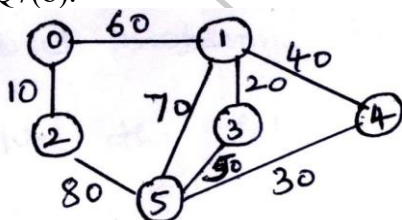


Fig.Q7(b)

OR

Q.8	a.	Solve the following single source shortest path problem assuming vertex '5' as the source. 	10	L3	CO4
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Fig.Q8(a)

- b. Write Huffman's algorithm. Construct Huffman tree and resulting code word for the following:

Character	A	B	C	D	E	-
Probability	0.5	0.35	0.5	0.1	0.4	0.2

Encode the text DAD_CBE.

Module – 5

Q.9	a.	Explain the following with example: (i) P problem (ii) NP problem	06	L1	CO5
	b.	What is decision tree? Construct decision tree for the three element insertion sort.	08	L2	CO5
	c.	Construct state space tree to solve 4 queens problem.	06	L3	CO5

OR

Q.10	a.	What is backtracking? Apply back tracking to solve the below instance of sum of subset problem: $s = \{3, 5, 6, 7\}$, $d = 15$	10	L3	CO6															
	b.	Solve the following instance of knapsack problem using branch and bound technique knapsack capacity = 10. <table><tr><td>Item</td><td>Weight</td><td>Value</td></tr><tr><td>1</td><td>4</td><td>40</td></tr><tr><td>2</td><td>7</td><td>42</td></tr><tr><td>3</td><td>5</td><td>25</td></tr><tr><td>4</td><td>3</td><td>12</td></tr></table>	Item	Weight	Value	1	4	40	2	7	42	3	5	25	4	3	12	10	L4	CO6
Item	Weight	Value																		
1	4	40																		
2	7	42																		
3	5	25																		
4	3	12																		

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Microcontroller

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the purpose of various fields of current program status register with a neat diagram.	05	L2	CO1
	b.	Explain the ARM design philosophy.	06	L2	CO1
	c.	Explain the core extensions of ARM processor with neat block diagram.	09	L2	CO1
OR					
Q.2	a.	Explain Embedded systems hardware with a neat block diagram.	06	L2	CO1
	b.	What is pipelines in ARM? Illustrate with an example the pipeline stage of ARM 9 and ARM 10.	09	L2	CO1
	c.	Describe the RISC design philosophy with 4 design rules.	05	L2	CO1
Module – 2					
Q.3	a.	Explain the following with examples : (i) RSC (ii) MLA (iii) STRH (iv) SWP	10	L2	CO2
	b.	Explain the different data processing instruction in ARM.	10	L2	CO2
OR					
Q.4	a.	Explain Barrel shifter instruction in ARM with suitable examples.	10	L2	CO2
	b.	Explain the different branch instruction of ARM processor.	05	L2	CO2
	c.	Explain co-processor instruction of ARM processor.	05	L2	CO2
Module – 3					
Q.5	a.	Explain the different basic data types in C. Provide examples of how each data type can be used in a C program.	08	L2	CO3
	b.	Discuss the concept of register allocation in compiler optimization. Illustrate its significance with an example.	07	L2	CO3
	c.	Describe the process of a function call in C.	05	L2	CO3
OR					
Q.6	a.	Discuss the common portability issues faced when writing C programs. How can these issues be mitigated.	07	L2	CO3
	b.	Explain the concept of pointer aliasing with example.	07	L2	CO3
	c.	How are function calls handled efficiently in calling function in C?	06	L2	CO3
Module – 4					
Q.7	a.	What are interrupts? Discuss interrupt vector table with diagram for ARM processor.	06	L2	CO4
	b.	Describe the sequence of operations that occurs when an ARM processor handles an IRQ exceptions.	06	L2	CO4
	c.	Discuss the priority system for exception in ARM processor.	08	L2	CO4
OR					
Q.8	a.	Explain the role of the link register in ARM exception handling.	08	L2	CO4
	b.	Explain the design and implementation of an interrupt stack in a ARM-based system. Explain the steps involved.	08	L2	CO4
	c.	What are the key differences between a boot loader and firmware?	04	L2	CO4

Module – 5					
Q.9	a.	Explain the basic operation of a cache controller.	06	L2	CO5
	b.	With a neat diagram, explain the basic architecture of a cache memory.	10	L2	CO5
	c.	Mention any 4 relationship between cache and main memory.	04	L2	CO5
OR					
Q.10	a.	Write a note on cache write policy both write back or write through.	10	L2	CO5
	b.	Describe the allocation policy on a cache miss.	04	L2	CO5
	c.	Write a note on following : (i) Write buffers (ii) Cache efficiency	06	L2	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Database Management System

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C																																									
Q.1	a.	Define the following terms: (i) Database (ii) Schema (iii) Entity (iv) DDL (v) Degree of a relationship	05	L1	CO1																																									
	b.	Briefly explain characteristics of database approach.	05	L2	CO1																																									
	c.	List and explain advantages of using DBMS approach.	10	L2	CO1																																									
OR																																														
Q.2	a.	Define the following terms: (i) Cardinality (ii) Weak entity (iii) Program data independence (iv) DML (v) Value sets	05	L1	CO1																																									
	b.	Describe three-schema architecture. Why do we need mappings between schema levels?	05	L2	CO1																																									
	c.	Explain different types of attributes in ER model with suitable example for each.	10	L2	CO1																																									
Module – 2																																														
Q.3	a.	With suitable example, explain the entity integrity and referential integrity constraints. Why each is considered important?	05	L2	CO2																																									
	b.	Discuss equijoin and natural join with suitable example using relational algebra notation.	05	L2	CO2																																									
	c.	Given the relational tables: <table><tr><td colspan="4">Employee:</td><td colspan="2">Department:</td></tr><tr><td>EID</td><td>Name</td><td>DepID</td><td>Salary</td><td>DeptID</td><td>DeptName</td></tr><tr><td>1</td><td>Alice</td><td>10</td><td>5000</td><td>10</td><td>HR</td></tr><tr><td>2</td><td>Bob</td><td>20</td><td>6000</td><td>20</td><td>IT</td></tr><tr><td>3</td><td>Eve</td><td>20</td><td>6500</td><td>30</td><td>Sales</td></tr></table> Project <table><tr><td>PID</td><td>Project Name</td><td>DeptID</td></tr><tr><td>101</td><td>Project Alpha</td><td>10</td></tr><tr><td>102</td><td>Project Beta</td><td>20</td></tr><tr><td>103</td><td>Project Gamma</td><td>30</td></tr></table> Write relational algebra expression for the following: (i) Find the names and salaries of all employees in the ‘IT’ department. (ii) Find the ID’s and names of employees who are in the ‘IT’ department and have a salary greater than 6000. (iii) Find the ID’s and names of employees who are either in the ‘HR’ department or have a salary greater than 6000. (iv) Find the names of employees who are not in the ‘IT’ department (v) Find the names of employees along with their department names.	Employee:				Department:		EID	Name	DepID	Salary	DeptID	DeptName	1	Alice	10	5000	10	HR	2	Bob	20	6000	20	IT	3	Eve	20	6500	30	Sales	PID	Project Name	DeptID	101	Project Alpha	10	102	Project Beta	20	103	Project Gamma	30	10	L3
Employee:				Department:																																										
EID	Name	DepID	Salary	DeptID	DeptName																																									
1	Alice	10	5000	10	HR																																									
2	Bob	20	6000	20	IT																																									
3	Eve	20	6500	30	Sales																																									
PID	Project Name	DeptID																																												
101	Project Alpha	10																																												
102	Project Beta	20																																												
103	Project Gamma	30																																												

1 of 3

OR

Q.4	a.	Explain any two operations that change the state of relation in a database. Provide suitable examples.	05	L2	CO2																																										
	b.	Discuss the aggregation functions and grouping in relational algebra with suitable examples.	05	L2	CO2																																										
	c.	Given the relational tables: <table><tr><th colspan="2">Student:</th></tr><tr><th>SID</th><th>Name</th></tr><tr><td>a</td><td>Alice</td></tr><tr><td>b</td><td>Bob</td></tr><tr><td>c</td><td>Carol</td></tr></table><table><tr><th colspan="2">Project:</th></tr><tr><th>PID</th><th>Project Name</th></tr><tr><td>p</td><td>Alpha</td></tr><tr><td>q</td><td>Beta</td></tr><tr><td>r</td><td>Gamma</td></tr></table> <table><tr><th colspan="2">Language:</th></tr><tr><th>LID</th><th>Language Name</th></tr><tr><td>x</td><td>Python</td></tr><tr><td>y</td><td>Java</td></tr><tr><td>z</td><td>C++</td></tr></table><table><tr><th colspan="2">Enrollment:</th></tr><tr><th>SID</th><th>PID</th></tr><tr><td>a</td><td>p</td></tr><tr><td>a</td><td>q</td></tr><tr><td>b</td><td>q</td></tr><tr><td>c</td><td>r</td></tr></table> Write relational algebra expression for the following: (i) Rename the student table to Learner and display it. (ii) Find the students (learners) who are not enrolled in any project. (iii) Find the students who are enrolled in all projects. (iv) Find the students who are not enrolled in any project. (v) Find the students who are enrolled in both the 'Alpha' and 'Beta' projects.	Student:		SID	Name	a	Alice	b	Bob	c	Carol	Project:		PID	Project Name	p	Alpha	q	Beta	r	Gamma	Language:		LID	Language Name	x	Python	y	Java	z	C++	Enrollment:		SID	PID	a	p	a	q	b	q	c	r	10	L3	CO2
Student:																																															
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r	Gamma																																														
Language:																																															
LID	Language Name																																														
x	Python																																														
y	Java																																														
z	C++																																														
Enrollment:																																															
SID	PID																																														
a	p																																														
a	q																																														
b	q																																														
c	r																																														

Module – 3

Q.5	a.	Explain Armstrong inference rules.	05	L2	CO4
	b.	What is the need for normalization? Explain 1NF, 2NF and 3NF with examples.	05	L2	CO4
	c.	What is functional dependency? Write an algorithm to find minimal cover for set of functional dependencies. Construct minimal cover M for set of functional dependencies which are: $E = \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$	10	L3	CO4

OR

Q.6	a.	Explain the types of update anomalies in SQL with an example.	05	L2	CO4
	b.	Explain types of JDBC drivers.	05	L2	CO5
	c.	Consider the schema $R = ABCD$, subjected to FDs $F = \{A \rightarrow B, B \rightarrow C\}$, and the non-binary partition $D1 = \{ACD, AB, BC\}$. State whether D1 is a lossless decomposition? [give all steps in detail].	10	L3	CO4

Module – 4

Q.7	a.	Define transaction. Discuss ACID properties.	05	L2	CO5
	b.	With a neat diagram, explain transition diagram of a transaction.	05	L2	CO5
	c.	Demonstrate working of assertion and triggers in SQL with example.	10	L3	CO5

OR

Q.8	a.	Explain cursor and its properties in embedded SQL with suitable example.	05	L2	CO5
	b.	Determine if the following schedule is serializable and explain your reasoning: i) $T1 : R(X)W(X) \quad T2 : R(X)W(X) \quad T1 : COMMIT \quad T2 : COMMIT$ ii) $T1 : W(X)R(Y) \quad T2 : R(X)W(Y) \quad T1 : COMMIT \quad T2 : COMMIT$	05	L2	CO5

	c.	Consider the tables below: Sailors (<u>sid</u> : integer, sname : string, rating : integer, age : real) Boats (<u>bid</u> : integer, bname : string, color : string); Reserves (<u>sid</u> : integer, <u>bid</u> : integer, day : date) Write SQL queries for the following: (i) Write create table statement for reserves. (ii) Find all information of sailors who have reserved boat number 101. (iii) Find the names of sailors who have reserved at least one boat. (iv) Find the names of sailors who have reserved a red boat. (v) Find the average age of sailors for each rating level.	10	L3	CO5
Module – 5					
Q.9	a.	Explain the CAP theorem.	05	L2	CO6
	b.	What is NOSQL graph database? Explain Neo4j.	05	L2	CO6
	c.	Why concurrency control and recovery are needed in DBMS? Demonstrate with suitable examples types of problems that may occur when two simple transactions run concurrently.	10	L3	CO5
OR					
Q.10	a.	Explain basic operations CRUD in MongoDB.	05	L2	CO6
	b.	Explain deadlock prevention protocols.	05	L2	CO5
	c.	Briefly discuss the two-phase locking techniques for concurrency control.	10	L3	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Discrete Mathematical Structures

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define a tautology. Prove that for any propositions p, q, r the compound propositions $\{(p \rightarrow q) \wedge (q \rightarrow r)\} \rightarrow (p \rightarrow r)$ is tautology.	06	L2	CO1
	b.	Establish the validity of the following argument using the rules of inference: $\{p \wedge (p \rightarrow q) \wedge (s \vee r) \wedge (r \rightarrow \sim q)\} \rightarrow (s \vee t)$	07	L2	CO1
	c.	For any two odd integers m and n, show that: (i) $m + n$ is even (ii) mn is odd	07	L2	CO1
OR					
Q.2	a.	Show that the compound proposition $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$ for primitive statements p, q, r is logically equivalent.	06	L2	CO1
	b.	Prove the following using law of logic: $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$	07	L2	CO1
	c.	Determine the truth value of each of the following quantified statements, the universe being the set of all non-zero integers: (i) $\exists x, \exists y, [xy = 1]$ (ii) $\exists x, \forall y, [xy = 1]$ (iii) $\forall x, \exists y, [xy = 1]$ (iv) $\exists x, \exists y, [(2x + y = 5) \wedge (x - 3y = -8)]$ (v) $\exists x, \exists y, [(3x - y = 17) \wedge (2x + 4y = 3)]$	07	L3	CO1
Module – 2					
Q.3	a.	Prove that for each $n \in \mathbb{Z}^+$, $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$.	06	L2	CO2
	b.	Let $a_0 = 1, a_1 = 2, a_2 = 3$ and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$ for $n \geq 3$, prove that $a_n \leq 3^n \forall n \in \mathbb{Z}^+$.	07	L2	CO2
	c.	How many positive integers n can be we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want n to exceed 5,000,000?	07	L3	CO2
OR					
Q.4	a.	By mathematical induction prove that $1.3 + 2.4 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$	06	L2	CO2
	b.	Find the number of permutations of the letters of the word ENGINEERING such that: (i) All the E's are together (ii) Arrangement begin with N (iii) All the vowels are adjacent.	07	L3	CO2
	c.	Find the coefficient of $a^2b^3c^2d^5$ in the expansion of $(a + 2b - 3c + 2d + 5)^{16}$.	07	L3	CO2
Module – 3					
Q.5	a.	State pigeon hole principle. Prove that if 30 dictionaries in a library contain a total of 61,327 pages then atleast one of the dictionaries must have atleast 2045 pages.	06	L3	CO3
	b.	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x-5 & \text{if } x > 0 \\ 1-3x & \text{if } x \leq 0 \end{cases}$. Find $f^{-1}(0), f^{-1}(1), f^{-1}(-1), f^{-1}(3), f^{-1}(-6), f^{-1}([-6, 5])$ and $f^{-1}([-5, 5])$	07	L2	CO3
	c.	Draw the Hasse diagram representing the positive divisor of 36.	07	L3	CO3

1 of 2

OR

Q.6	a.	Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$, (i) How many functions are there from A to B? (ii) How many of these are one to one? (iii) How many functions are there from B to A? (iv) How many of these are onto?	06	L2	CO3
	b.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$. If $(g \circ f)(x) = 9x^2 - 9x + 3$, determine a and b .	07	L2	CO3
	c.	Let $A = \{1, 2, 3, 4, 6\}$ and R be a relation on A defined by aRb if and only if “ a is multiple of b ”. Write down the relation R , relation matrix $M(R)$ and draw the digraph. List out in degree and out degree.	07	L3	CO3

Module – 4

Q.7	a.	In how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block?	06	L3	CO4
	b.	Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2, 3, or 5.	07	L3	CO4
	c.	Solve the recurrence relation $a_{n+2} - 3a_{n+1} + 2a_n = 0$, $a_0 = 1$, $a_1 = 6$.	07	L2	CO4

OR

Q.8	a.	In how many ways can the 26 letters of the English alphabet be permuted so that none of the patterns CAR, DOG, PUN or BYTE occurs?	06	L3	CO4
	b.	Five teachers T_1, T_2, T_3, T_4 are to be made class teachers for five classes, C_1, C_2, C_3, C_4, C_5 , one teacher for each class. T_1 and T_2 do not wish to become the class teachers for C_1 or C_2 , T_3 and T_4 for C_4 or C_5 , and T_5 for C_3 or C_4 or C_5 . In how many ways can the teachers be assigned the work? (Without displeasing any teacher)	07	L3	CO4
	c.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ where $n \geq 0$ and $F_0 = 0$, $F_1 = 1$.	07	L2	CO4

Module – 5

Q.9	a.	If G be a set of all non zero real numbers and let $a * b = \frac{ab}{2}$ then show that $(G, *)$ is an abelian group.	06	L2	CO5
	b.	Define Klein group and if $A = \{e, a, b, c\}$ then show that this is a Klein-4 group.	07	L2	CO5
	c.	State and prove Lagrange's theorem.	07	L2	CO5

OR

Q.10	a.	If H and K are subgroups of group G, prove that $H \cap K$ is also a subgroup of G. Is $H \cup K$ a subgroup of G?	06	L2	CO5																																																	
	b.	Define cyclic group and show that $(G, *)$ whose multiplication table is as given below is cyclic. <table border="1"><tr><td>*</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>a</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>b</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td></tr><tr><td>c</td><td>c</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td></tr><tr><td>d</td><td>d</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td></tr><tr><td>e</td><td>e</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>f</td><td>f</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr></table>	*	a	b	c	d	e	f	a	a	b	c	d	e	f	b	b	c	d	e	f	a	c	c	d	e	f	a	b	d	d	e	f	a	b	c	e	e	f	a	b	c	d	f	f	a	b	c	d	e	07	L2	CO5
*	a	b	c	d	e	f																																																
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	c.	Let $G = S_4$, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup $H = \langle \alpha \rangle$. Determine the left cosets of H in G.	07	L3	CO5																																																	

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Biology for Engineers (CSE)

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M*: Marks, *L*: Bloom's level, *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is stem cell? Explain its types and list its applications.	7	L2	CO1
	b.	Explain in detail the properties and functions of nucleic acids.	6	L2	CO1
	c.	Explain the importance of special biomolecules.	7	L2	CO1
OR					
Q.2	a.	What is a biomolecule? Explain the classifications of biomolecule.	7	L2	CO1
	b.	Explain the properties and functions of carbohydrates.	6	L2	CO1
	c.	Describe the structure and functions of a cell with a neat diagram.	7	L3	CO1
Module – 2					
Q.3	a.	What is the role of lipids? Outline the process of obtaining biodiesel from lipids.	7	L3	CO2
	b.	Differentiate between PHA and PLA as a bioplastic materials.	6	L4	CO1
	c.	Explain the role of DNA vaccine for rabies and RNA vaccine for COVID-19.	7	L2	CO1
OR					
Q.4	a.	What are the key properties, advantages and limitations of cellulose based water filters.	7	L3	CO2
	b.	How can DNA finger printing be applied to evaluate its effectiveness and reliability in forensic applications.	6	L4	CO1
	c.	Describe the use of meat analogue and plant protein as food.	7	L2	CO2
Module – 3					
Q.5	a.	Deliberate the functioning of brain as CPU system.	7	L3	CO2
	b.	Write a short note on spirometry and ventilator.	6	L2	CO2
	c.	Explain heart as pump system.	7	L3	CO2

1 of 2

BBOC407					
OR					
Q.6	a.	Explain eye as a camera system.	7	L3	CO2
	b.	Write a short note on cardiac pacemaker.	6	L2	CO2
	c.	Explain kidney as purification system.	7	L3	CO2
Module – 4					
Q.7	a.	Describe the materials used and engineering applications of Velcro technology.	7	L3	CO3
	b.	Compare the process of photosynthesis to the functioning of photovoltaic cells.	6	L4	CO3
	c.	Explain the HBOCs and PFCs as human blood substituents.	7	L3	CO3
OR					
Q.8	a.	Explain the terms lotus leaf effect and bird flying.	7	L3	CO3
	b.	Compare biological echolocation and technological echolocation highlighting their applications in navigation and detection.	6	L4	CO3
	c.	Explain the terms shark skin, swim suits and bullet train using biological concepts.	7	L3	CO3
Module – 5					
Q.9	a.	Compare the functioning of electrical tongue and human tongue.	7	L4	CO4
	b.	Explain muscle cells as scaffold for tissue growth.	6	L2	CO4
	c.	Explain bioremediation and biomining via microbial surface adsorption.	7	L2	CO4
OR					
Q.10	a.	Illustrate the basic steps of bioprinting process and list the various types of bioprinting techniques.	7	L4	CO4
	b.	Write a short note on: i) Importance of DNA origami ii) Self healing bioconcrete.	6	L2	CO4
	c.	Discuss the applications of artificial intelligence in the diagnosis of disease.	7	L2	CO4

GBCS SCHEME

BUHK408

USN

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

1. What is the state of liking and a holistic and all-encompassing state of the mind that crates inner harmony?
a) Prosperity b) Happiness c) Innateness d) Organized
2. _____ is defined as a group of people working together to create a surplus
a) Alone b) Organisation c) System d) Army
3. The process whereby national economies and business systems are becoming deeply interlinked with each other is called
a) Nationalization b) Glocalization c) Global linking d) Globalization
4. Which of the following actions will likely lead to organizational ethical behavior?
a) Establishing an ethics office
b) Promoting moral courage
c) Developing strong governance processes
d) All of these
5. _____ behavior tends to arise when managers decide to put the attainment of their own personal goals, or the goals of the organization, above the fundamental rights of one or more stakeholder groups
a) Complementary b) Situational c) Unethical d) Confusing
6. Accepted principles of right or wrong governing the conduct of business people are called :
Choose one answer
a) Business b) Conduct c) Business ethics d) Principles
7. What is true about a professional
a) The professional has to have autonomy
b) He cannot be supervised by the client
c) He should not subject to political control
d) All of these

8. The basic rule of professionalism is “not knowingly to do _____”
a) Harm b) Work c) Competition d) Copying
9. The study of _____, as contrasted with legal, obligations is called ethics
a) Behaviors b) Moral c) Business d) All of these
10. Ethics is a conception of _____conduct
a) Bad/Wrong b) Good/Correct c) Right/True d) Right and Wrong
11. Ethics and _____go together
a) Profits b) Management c) Worker ship d) All of these
12. _____driven companies are always successful in the long run
a) Management b) Worker union c) Value d) Profit
13. A business needs to be ethical, in order to
a) Fulfill public expectations for business
b) Prevent harming others
c) Improve business relations and employees productivity
d) All of these
14. Following is (are) types(s) of ethics issues(s)
a) Functional area ethics b) Corporate policy ethics
c) Face to face ethics d) All of these
15. What is called living with assumption for oneself as body and living of human being only on the basis of physical facilities, and not with right understanding and relationship?
a) Human consciousness b) Happiness
c) Right Understanding d) Animal Consciousness
16. Samridhi means
a) Happiness b) Wealth c) Prosperity d) Health
17. Developed nations are the live example of
a) Prosperity b) Wealth c) Happiness d) Health
18. The participation of human beings is seen in two forms
a) Prosperity and work b) Values and Understanding
c) Behavior and Wealth d) Behavior and Work
19. What are the outcomes of realization and understanding?
a) Work b) Values c) Happiness d) Health
20. Corporate culture _____ refers to how widely and deeply employees hold the company’s dominant values and assumptions
a) Breadth b) Depth c) Character d) Strength
21. The observable symbols and signs of a organization’s culture are called :
a) Its cultural design b) Artifacts c) Cultural formatting d) Its cultural dcor
22. _____is the feeling or responsibility and commitment for ensuring Right Understanding and Right Feeling in the self of my relative
a) Guidance b) Care c) Respect d) Affection
23. The ability to perform a specific task, action, or function successfully is called _____
a) Competence b) Purpose c) Intention d) Program

24. To live in harmony with other human beings is _____
a) Svatva b) Swatantrata c) Swarajya d) None of these
25. Natural acceptance is?
a) Universal b) Definite c) Continuous d) All of these
26. Basic human aspirations are _____
a) Happiness b) Prosperity c) Education d) Both (a) and (b)
27. Which of the following is true?
a) Happiness is definite and it can be understood
b) I need to be unhappy to recognize that I am happy
c) One will be bored of happiness if one is always happy
d) I will grow only if I am unhappy. If I become happy my growth will stop
28. To be wealthy is _____ condition in the modern world to be happy
a) Not essential b) True c) Fake d) The most important
29. It is important to have a good _____ to be successful in your career
a) Appear b) Attend c) Attitude d) Respectful
30. _____ skills such as writing, speaking and body language are important work ethics so that you can be clear in your messages to others
a) Communication b) Cooperation c) Honesty d) Integrity
31. You are a person with a high level of _____ if you stand by your values even if others put pressure on you to do something that is wrong or if they make fun of you
a) Communication b) Cooperation c) Honesty d) Integrity
32. If you don't have _____ you can really get in trouble for lying, cheating or deceiving others
a) Communication b) Cooperation c) Honesty d) Integrity
33. If you have _____ you will be a trusted person because they will see that you are committed to your company
a) Loyalty b) Organizational Skills c) Productivity d) Respect
34. Some jobs cannot be done alone and some can be done much better and faster if you possess the quality of _____
a) Responsibility b) Respect c) Teamwork d) Cooperation
35. The three major types of ethical issues include except?
a) Communication issues b) Systematic issues
c) Corporate issues d) Individual issues
36. Ethics are moral principles and values which _____
a) Guide a firm's behavior
b) Govern the actions of an individual
c) Provide employees with rules on how to behave
d) Are legally enforceable
37. The word "Ethics" comes from
a) Ethos b) Eternity c) Elementary d) Essentiality
38. A _____ is a set of principles that people hold themselves to or that are held by organizations or groups
a) Code of ethics b) Code of honor c) Ethical perspective d) Persuasive tactic

39. _____ are beliefs and attitudes we have that can actually conflict with our ethical decision
a) Morals b) Principles c) Virtues d) Values
40. Intellectual property rights are an important aspect in international trade. Which of the following is covered under IPR?
a) Patents b) Copyrights c) Trademarks d) All of these
41. Which of the following is not the infringement of copyright?
a) Copy the software to another computer by educational institution
b) Copy the software to another computer by company
c) Copy the software to another computer
d) To make backup copies
42. Patent can be granted for _____?
a) Process b) Ideas
c) Both product and process d) Product
43. Government may order the non advertisement of any patent application in the case of invention related to _____
a) Information technology b) Food
c) Drug d) Defense technology
44. Patent for the _____ is prohibited
a) Traditional knowledge b) Arms
c) Atomic energy d) Junk food
45. A design can be also registered under _____
a) Trademark Act b) Patent Act
c) Geographical Indication Act d) Copyright Act
46. IPR Complete specification gives _____
a) Full description of claim only
b) Full description of royalty
c) Full description of invention
d) Full description of application
47. Is the first owner of a Book's copyright?
a) Author b) Publisher c) Translator d) Printing press owner
48. Copyright is a _____
a) Creation by person of efforts, intellectual and capital
b) Creation by person of the efforts only
c) Creation by negligence of a person
d) Creation by the mental status of a person
49. Globalization has improved the living structure of which of the following?
a) All the people
b) People living in developing countries
c) People living in developed countries
d) None of these
50. Which sector experienced the least profit due to the effect of Globalization?
a) Industrial sector b) Service sector c) Agriculture Sector d) Secondary sector

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain software process and software engineering practices.	10	L2	CO1
	b.	Explain the waterfall model and incremental model, with diagram.	10	L2	CO1
OR					
Q.2	a.	Explain Boehm Spiral process model with a neat diagram. Mention its advantages and disadvantages.	10	L2	CO1
	b.	Explain the five activities of a generic process framework for software engineering.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the distinct tasks of requirement engineering.	10	L2	CO2
	b.	Illustrate the UML use case diagram for safe home system.	10	L2	CO2
OR					
Q.4	a.	Explain Class-Responsibility-Collaborator(CRC) modeling and data modeling with an example.	10	L2	CO2
	b.	Explain the elements of analysis model in requirement modeling.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the principles of agile process development.	10	L2	CO3
	b.	Explain the following : i) Adaptive software development ii) SCRUM	10	L2	CO3
OR					
Q.6	a.	Explain the concepts of extremes programming with a neat diagram.	10	L2	CO3
	b.	Explain design modeling principles that guide the respective framework activity.	10	L2	CO3
Module – 4					
Q.7	a.	Illustrate the project management life cycle with a neat diagram.	10	L2	CO4
	b.	Explain : i) Different ways of categorizing software projects ii) Smart objectives	10	L2	CO4
OR					
Q.8	a.	Explain the difference between traditional versus modern project management practices along with the role of management.	10	L3	CO4
	b.	Explain software development life cycle (ISO 12207) with a neat diagram.	10	L2	CO4
Module – 5					
Q.9	a.	Explain Quality Management System with principles of BS EN ISO-9001-2000.	10	L2	CO5
	b.	Explain the following : i) McCall model ii) Garvin's Quality Dimensions.	10	L2	CO5
OR					
Q.10	a.	Describe six generic functions allowed in automated estimation techniques of software projects.	10	L3	CO5
	b.	Explain COCOMO II model.	10	L2	CO5

CBCS SCHEME

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BCS502

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Computer Networks

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is data communication? List and explain characteristics and components of communication model.	06	L1	CO1
	b.	Define switching. Explain Circuit Switched Network and Packet Switched Network.	06	L2	CO1
	c.	With neat sketch, explain different layers of TCP/IP protocol suite.	08	L2	CO1
OR					
Q.2	a.	What are guided transmission media? Explain twisted pair cable in detail.	06	L1	CO1
	b.	What is Virtual Circuit Network (VCN)? With neat diagram, explain three phases involved in VCN.	08	L1	CO1
	c.	Write a note on Encapsulation and decapsulation at Source Host for TCP/IP protocol suite.	06	L2	CO1
Module – 2					
Q.3	a.	Define Redundancy. Explain CRC encoder and CRC decoder operation with block diagram.	08	L2	CO2
	b.	Distinguish between Flow Control and Error Control. Explain Stop and Wait Protocol.	08	L2	CO2
	c.	List and explain Control Fields of I-frames, S-frames and U-frames.	04	L2	CO2
OR					
Q.4	a.	What is Hamming distance? With example, explain Parity Check Code.	06	L1	CO2
	b.	Define Framing. Explain character oriented framing and bit-oriented framing.	06	L1	CO2
	c.	With flow diagram, explain CSMA/CA.	08	L2	CO2
Module – 3					
Q.5	a.	Explain virtual-circuit approach to route the packets in packet-switched network.	10	L2	CO3
	b.	Illustrate the working of OSPF and BGP.	10	L3	CO3
OR					
Q.6	a.	Explain IPv6 datagram format.	10	L2	CO3
	b.	Write an Dijkstra's algorithm to compute shortest path through graph.	06	L1	CO3
	c.	Write a note on Routing Information Protocol (RIP) algorithm.	04	L2	CO3
Module – 4					
Q.7	a.	Explain Go-Back-N protocol working.	10	L2	CO4
	b.	With neat sketch, explain three-way handshaking of TCP connection establishment.	10	L2	CO4

OR

Q.8	a.	With an outline, explain selective repeat protocol.	10	L2	CO4
	b.	List and explain various services provided by User Datagram Protocol (UDP).	10	L2	CO4

Module – 5

Q.9	a.	Briefly explain Secure Shell (SSH).	10	L2	CO4
	b.	Write a note on Request message and response message formats of HTTP.	10	L2	CO4

OR

Q.10	a.	With neat diagram, explain the basic model of FTP.	04	L2	CO4
	b.	Describe the architecture of electronic mail (e-mail).	06	L3	CO4
	c.	Briefly explain Recursive Resolution and Iterative Resolution in DNS.	10	L2	CO4

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Theory of Computation

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C														
Q.1	a.	Define the following with example : i) Language ii) String iii) Power of an alphabet.	3	L1	CO1															
	b.	Define DFA. Draw a DFA to accepts. i) The set of all strings that contain a substring aba. ii) To accept the strings of a's and b's that contain not more than three b's. iii) $L = \{w \in \{a, b\}^* : \text{No 2 consecutive characters are same in } w\}$.	10	L3	CO1															
	c.	Convert the following NFA to DFA. <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→ p</td><td>{p, q}</td><td>{p}</td></tr><tr><td>q</td><td>{r}</td><td>{r}</td></tr><tr><td>r</td><td>{s}</td><td>ϕ</td></tr><tr><td>* s</td><td>{s}</td><td>{s}</td></tr></table>		0	1	→ p	{p, q}	{p}	q	{r}	{r}	r	{s}	ϕ	* s	{s}	{s}	7	L2	CO1
	0	1																		
→ p	{p, q}	{p}																		
q	{r}	{r}																		
r	{s}	ϕ																		
* s	{s}	{s}																		
OR																				
Q.2	a.	Define the following with example : i) Alphabet ii) Reversal of string iii) Concatenation of Languages.	3	L1	CO1															
	b.	Design a DFA for the Language : $L = \{w \in \{0, 1\}^* : w \text{ is a string divisible by } 5\}$.	7	L3	CO1															
	c.	Define NFA. Obtain an ϵ - NFA which accepts strings consisting of 0 or more a's , followed by 0 or more b's followed by 0 or more C's. Also convert it to DFA.	10	L2	CO1															
Module – 2																				
Q.3	a.	Define Regular expression. Write the regular expression for the following languages : i) Strings of a's and b's starting with a and ending with b. ii) Set of strings that consists of alternating 0's and 1's. iii) $L = \{a^n b^m , (n + m) \text{ is even}\}$. iv) $L = \{w : w \bmod 3 = 0 , \text{ where } w \in \{a, b\}^*\}$.	10	L2	CO2															

	b.	Minimize the following finite automata using Table filling algorithm : <table><tr><th>δ</th><th>a</th><th>b</th></tr><tr><td>$\rightarrow$ A</td><td>B</td><td>A</td></tr><tr><td>B</td><td>A</td><td>C</td></tr><tr><td>C</td><td>D</td><td>B</td></tr><tr><td>*</td><td>D</td><td>A</td></tr><tr><td>E</td><td>D</td><td>F</td></tr><tr><td>F</td><td>G</td><td>E</td></tr><tr><td>G</td><td>F</td><td>G</td></tr><tr><td>H</td><td>G</td><td>D</td></tr></table>	δ	a	b	$\rightarrow$ A	B	A	B	A	C	C	D	B	*	D	A	E	D	F	F	G	E	G	F	G	H	G	D	10	L2	CO2
δ	a	b																														
$\rightarrow$ A	B	A																														
B	A	C																														
C	D	B																														
*	D	A																														
E	D	F																														
F	G	E																														
G	F	G																														
H	G	D																														
OR																																
Q.4	a.	Construct ε - NFA for the following Regular expression : i) $(0 + 1) 0 1(1 + 0)$ ii) $1(0 + 1)^* 0$ iii) $(0 + 1)^* 0 1 1^*$	6	L1	CO2																											
	b.	Obtain the Regular expression that denotes the language accepted by Fig. Q4(b). Fig. Q4(b) Using Kleene's theorem.	6	L3	CO2																											
	c.	State the Pumping Lemma for the Regular Languages. And also prove that the following languages are not regular. i) $L = \{0^n 1^m \mid n \leq m\}$ ii) $L = \{0^n 1^m 2^n \mid n, m \geq 1\}$.	8	L1	CO2																											
Module – 3																																
Q.5	a.	Design CFG for the following languages : i) $L = \{a^n b^{n+3}, n \geq 0\}$ ii) $L = \{a^i b^j c^k, j = i + k, i \geq 0, k \geq 0\}$ iii) $L = \{w \mid w \bmod 3 > 0 \text{ where } w \in \{a\}^*\}$ iv) $L = \{a^m b^n \mid m \neq n\}$ v) Palindromes over 0 and 1.	10	L3	CO3																											
	b.	Consider the grammar G with productions. $S \rightarrow A b B / A / B$; $A \rightarrow aA / \varepsilon$; $B \rightarrow a B / b B / \varepsilon$. Obtain LMD, RMD and parse tree for the string aabab. Is the given grammar ambiguous?	10	L2	CO3																											
OR																																
Q.6	a.	Define the following with example : i) Context free grammar ii) Left most Derivation iii) Parse tree iv) Ambiguous grammar.	4	L1	CO3																											
	b.	Design PDA for the language : $L = \{a^i b^j c^k \mid i + k = j, i \geq 0, k \geq 0\}$ and show the moves made by the PDA for the string aabbbc.	10	L3	CO3																											

	c.	Convert the following CFG's to PDA : $S \rightarrow a A$; $A \rightarrow a ABC / bB / a$; $B \rightarrow b$; $C \rightarrow c$.	6	L2	CO3
Module – 4					
Q.7	a.	Define CNF. Convert the following CFG to CNF $E \rightarrow E + T / T$ $T \rightarrow T * F / F$ $F \rightarrow (E) / I$ $I \rightarrow Ia / Ib / a / b$.	10	L2	CO4
	b.	Show that $L = \{0^n 1^n 2^n / n \geq 1\}$ is not context free.	4	L2	CO4
	c.	Prove that the family of context free languages is closed under union and concatenation.	6	L1	CO4
OR					
Q.8	a.	Define Greibach Normal Form. Convert the following CFG to GNF. $S \rightarrow AB$; $A \rightarrow aA / bB / b$; $B \rightarrow b$.	6	L2	CO4
	b.	Consider the following CFG : $S \rightarrow ABC / BaB$ $A \rightarrow aA / BaC / aaa$ $B \rightarrow bBb / a / D$ $C \rightarrow CA / AC$ $D \rightarrow \epsilon$ i) What are useless symbols? ii) Eliminate ϵ - productions , Unit productions and useless symbols from the grammar.	10	L3	CO4
	c.	Prove that the following languages are not context free. i) $L = \{a^i / i \text{ is prime}\}$ ii) $L = \{a^n / n \geq 1\}$.	4	L2	CO3
Module – 5					
Q.9	a.	Define a Turing machine and explain with neat diagram, the working of a basic Turing machine.	6	L1	CO4
	b.	Design a Turing machine to accept the language, $L = \{a^n b^n c^n / n \geq 1\}$. Draw the transition diagram and show the moves for the string aabbcc.	14	L4	CO4
OR					
Q.10	a.	Design a Turing machine to accept palindrome over $\{a, b\}$ and draw the transition diagram.	12	L4	CO5
	b.	Write a short notes on : i) Recursively Enumerable Language. ii) Multitape Turing Machine.	8	L1	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
 2. The ratio between energy flows at different points in a food chain is known as
a) Ecological capacity b) Ecological efficiency
c) Ecological assimilation d) Ecological potential
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a) An animal that is fed upon another animal
b) Animal that feeds upon both plants and animals
c) An animal that feeds upon another animal
d) A primary consumer
 4. Why Rann of Kutch attracts aquatic birds in monsoon season?
a) Because desert land is converted to forest land
b) Because desert land is converted to snow
c) Because desert land do not convert
d) Because desert land is converted to salt marshes
 5. Which kind of soil we can find on the surface of Thar desert?
a) Rocky b) Moist c) Fertile d) Aeolian
 6. Which of the following type of forest important for watersheds?
a) Tropical Evergreen forests b) Tropical Deciduous forests
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7. Hot spots areas have
 - a) Low density of biodiversity
 - b) Only endangered plants
 - c) High density of hot springs
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 - a) meeting present needs without compromising on future needs
 - b) progress of human beings
 - c) balance between human needs and the ability of earth to provide the resources
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11. Veld type grasslands are located at
 - a) South Africa
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 - c) Australia
 - d) Britain
12. Which pyramid is always upright?
 - a) Energy
 - b) Biomass
 - c) Numbers
 - d) Food chain
13. In what form is solar energy is radiated from the Sun?
 - a) Ultraviolet Radiation
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14. What does MHD stands for in the energy field?
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15. The 'Miracle Material' that can turn CO₂ into liquid fuel is :
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 - a) Converts heat energy to chemical energy
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 - c) Converts chemical energy to electrical energy
 - d) Converts kinetic energy to heat energy

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 a) Section 12 b) Section 15 c) Section 18 d) Section 19
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32. Which among the following is used to dump the waste collected in the cities?
 a) Land fills b) Ocean c) River d) All of these

45. Preparation of Guidelines for Environmentally sound Management of e – waste is a duty assigned to
a) Producer b) Consumer c) MOEFCC d) SP CB/PCC
46. What is India's global rank in e – waste?
a) 3 b) 13 c) 23 d) 33
47. When did the Karnataka State Pollution Control Board for prevention and control of water pollution constituted?
a) 1974 b) 1978 c) 1982 d) 1986
48. Aerosol consisting of liquid droplets is called as
a) Mist b) Dust c) Fog d) Aerosol
49. Which of the following is non – point source of water pollution?
a) Factories b) Sewage treatment plant
c) Urban and suburban land d) All of these
50. When is World Water day celebrated?
a) January 26th b) June 5th c) September 22nd d) March 22nd

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

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1. What is Extended Producer Responsibility (EPR) as per the e – waste management rules in India?
 - a) The responsibility of consumer to manage e – waste
 - b) The responsibility of manufactures to manage e – waste throughout the product life cycle
 - c) The responsibility of retailers to manage e – waste disposal
 - d) The responsibility of informal recyclers to manage e – waste.
 2. Which international agreement regulates the transboundary movements of hazardous waste, including e – waste?
 - a) Kyoto Protocol
 - b) Paris agreement
 - c) Montreal Protocol
 - d) Basel convention
 3. Which colour bin is used for e – waste?
 - a) Blue
 - b) Green
 - c) Yellow
 - d) Black
 4. What are the health hazards which can be caused by E – waste?
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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

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- a) Genetic diversity
 - b) Community and ecosystem diversity
 - c) Species diversity within a community or ecosystem
 - d) Diversity among the plant
50. Algae , green plants and photosynthetic bacteria are
- a) Autotrophic
 - b) Heterotrophic
 - c) Decomposers
 - d) Consumers

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies and E – Waste Management

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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-
1. What is the Dissolved oxygen value required for the survival of aquatic species?
a) 7 mg/L b) 8.2 mg/L c) 6.5 mg/L d) 4 mg/L
 2. Which among the following is used to dump the waste collected in the cities?
a) Land fills b) Ocean c) River d) All of these
 3. Which type of waste includes items such as leftover food, fruit peels and yard trimmings?
a) Hazardous waste b) Organic waste
c) Bio – medical waste d) Electronic waste
 4. Which of the integrated waste management is reduced on an individual level?
a) Source Reduction b) Recycling c) Disposal d) Burning
 5. What is called for the process of burning municipal solid waste in a properly designed furnace under suitable temperature and operating conditions?
a) Landfill b) Recycling c) Vermicomposting d) Incineration
 6. The process of decomposition of biodegradable solid waste by earthworms is called
a) Landfill b) Vermicomposting c) Composting d) Shredding
 7. _____ is a liquid that passes through solid waste and extracts suspended impurities from it
a) Leachate b) Sludge c) Distilled water d) Municipal
 8. The colour code of plastic bag for disposing of microbial laboratory culture waste
a) black b) red c) blue d) white
 9. Average hospital waste produced per bed per day in Government hospital is
a) 1.5 to 2 kg b) 0.5 – 4 kg c) 0.5 to 1 kg d) 0.5 – 2 kg

10. Which of the following are the main contributors of the e –waste in the world?
 - I. Refrigerators / freezers , washing machines , dishwashers.
 - II. Small household appliances
 - III. Personal computers, telephones , laptops , printers.
 - IV. Gas cylinders, chimneys and home appliances
 - a) Only I, II, III b) Only I & II c) Only I, III , IV d) All of these
11. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
 - a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere
12. The ratio between energy flows at different points in a food chain is known as
 - a) Ecological capacity b) Ecological efficiency
 - c) Ecological assimilation d) Ecological potential
13. A predator is
 - a) An animal that is fed upon another animal
 - b) Animal that feeds upon both plants and animals
 - c) An animal that feeds upon another animal
 - d) A primary consumer
14. Why Rann of Kutch attracts aquatic birds in monsoon season?
 - a) Because desert land is converted to forest land
 - b) Because desert land is converted to snow
 - c) Because desert land do not convert
 - d) Because desert land is converted to salt marshes
15. Which kind of soil we can find on the surface of Thar desert?
 - a) Rocky b) Moist c) Fertile d) Aeolian
16. Which of the following type of forest important for watersheds?
 - a) Tropical Evergreen forests b) Tropical Deciduous forests
 - c) Tropical Montana forests d) Grassland forest
17. Hot spots areas have
 - a) Low density of biodiversity b) Only endangered plants
 - c) High density of hot springs d) High density of biodiversity
18. Sustainable Development means
 - a) meeting present needs without compromising on future needs
 - b) progress of human beings
 - c) balance between human needs and the ability of earth to provide the resources
 - d) all of these
19. The term Alpha diversity refers to
 - a) Genetic diversity b) Community and ecosystem diversity
 - c) Species diversity within a community or ecosystem
 - d) Diversity among the plant
20. Algae , green plants and photosynthetic bacteria are
 - a) Autotrophic b) Heterotrophic c) Decomposers d) Consumers

21. What is Extended Producer Responsibility (EPR) as per the e – waste management rules in India?
- The responsibility of consumer to manage e – waste
 - The responsibility of manufactures to manage e – waste throughout the product life cycle
 - The responsibility of retailers to manage e – waste disposal
 - The responsibility of informal recyclers to manage e – waste.
22. Which international agreement regulates the transboundary movements of hazardous waste, including e – waste?
- Kyoto Protocol
 - Paris agreement
 - Montreal Protocol
 - Basel convention
23. Which colour bin is used for e – waste?
- Blue
 - Green
 - Yellow
 - Black
24. What are the health hazards which can be caused by E – waste?
- Lung cancer
 - DNA damage
 - Brain
 - All of these
25. Preparation of Guidelines for Environmentally sound Management of e – waste is a duty assigned to
- Producer
 - Consumer
 - MOEFCC
 - SP CB/PCC
26. What is India's global rank in e – waste?
- 3
 - 13
 - 23
 - 33
27. When did the Karnataka State Pollution Control Board for prevention and control of water pollution constituted?
- 1974
 - 1978
 - 1982
 - 1986
28. Aerosol consisting of liquid droplets is called as
- Mist
 - Dust
 - Fog
 - Aerosol
29. Which of the following is non – point source of water pollution?
- Factories
 - Sewage treatment plant
 - Urban and suburban land
 - All of these
30. When is World Water day celebrated?
- January 26th
 - June 5th
 - September 22nd
 - March 22nd
31. _____ is caused by drinking water high in nitrates.
- Cholera
 - Kidney problem
 - Liver problem
 - Methomoglobinemia
32. Bhopal gas tragedy took place in the year _____ and the gas responsible was _____
- 1964, Hydrogen fluoride
 - 1974, Methyl chloride
 - 1984, methyl ISO – cyanide
 - 1994, Methyl sulphate
33. The major chemical pollutants in photochemical smog are
- NO, NO₂, VOC, O₃, PAN
 - N₂O, NO₂, VOC, O₃, PAN
 - NO, NO₂, VOC, O₂, PAN
 - NO, N₂O₅, VOC, O₃, PAN

34. The international protocol to protect the Ozone layer is
 - a) Vienna protocol
 - b) Kyoto protocol
 - c) Cartagena protocol
 - d) Montreal protocol
35. Which is the best and the worst method of plume behavior for pollution dispersion?
 - a) Lofting and fumigation
 - b) Trapping and fanning
 - c) Conning and fumigation
 - d) Fanning and Lofting
36. What is called for a Temporary hearing loss?
 - a) Temporary ear pain
 - b) Temporary hearing problem
 - c) Temporary threshold shift
 - d) Temporary hearing shift
37. What timings loud speakers shouldn't use in public areas?
 - a) 10.00 pm to 5.00 am
 - b) 11.00 pm to 6.00 am
 - c) 1.00 am to 7.00 am
 - d) 10.00 pm to 6.00 am
38. In which section, if a person violates the noise pollution regulations , is liable for penalty according to Environmental Protection Act, 1986.
 - a) Section 12
 - b) Section 15
 - c) Section 18
 - d) Section 19
39. 'Minamata Disease' is caused due to
 - a) Lead
 - b) Arsenic
 - c) Mercury
 - d) Cadmium
40. The process of reducing the fluoride content from water is called as
 - a) Chlorination
 - b) Fluoridation
 - c) Defluoridation
 - d) Fanning and Lofting
41. Veld type grasslands are located at
 - a) South Africa
 - b) South America
 - c) Australia
 - d) Britain
42. Which pyramid is always upright?
 - a) Energy
 - b) Biomass
 - c) Numbers
 - d) Food chain
43. In what form is solar energy is radiated from the Sun?
 - a) Ultraviolet Radiation
 - b) Infrared Radiation
 - c) Electromagnetic waves
 - d) Transverse waves
44. What does MHD stands for in the energy field?
 - a) Magneto Hydro Dynamic
 - b) Metal Hydrogen Detox
 - c) Micro Hybrid Drive
 - d) Metering Head Differential
45. The 'Miracle Material' that can turn CO₂ into liquid fuel is :
 - a) Propane
 - b) Copper
 - c) Graphene
 - d) Potassium
46. A tide whose difference between high and low tide is greatest.
 - a) Diurnal tide
 - b) Neap tide
 - c) Spring tide
 - d) Ebb tide
47. Which of the turbine can be mounted vertically and horizontally.
 - a) Pelton wheel
 - b) Kaplan turbine
 - c) Gorlov turbine
 - d) Francis turbine

48. Which type of fuel is removed from the reactor core after reaching end of core life service?
- a) Burnt fuel b) Spent fuel c) Engine oil d) Radioactive fuel
49. What is a fuel cell?
- a) Converts heat energy to chemical energy
b) Converts heat energy to electrical energy
c) Converts chemical energy to electrical energy
d) Converts kinetic energy to heat energy
50. Which one of the following is the apex organization in our country in the field of pollution control?
- a) Water Pollution Control Board b) State Pollution Control Board
c) Central Pollution Control Board d) Air Pollution Control Board

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

UNIX System Programming

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the Kernel and Shell relationship in UNIX operating system with a neat diagram.	10	L1	CO1
	b.	Explain the following UNIX commands with syntax and examples: i) who ii) ls iii) passwd iv) echo v) date	10	L2	CO1
OR					
Q.2	a.	Explain any five file related commands with syntax and example of each.	10	L2	CO1
	b.	Explain the salient features of UNIX operating system.	04	L1	CO1
	c.	Explain the file types or categories.	06	L2	CO1
Module – 2					
Q.3	a.	Explain the use of chmod command to change file permission using both absolute and relative methods.	10	L2	CO2
	b.	Explain ls commands with all the options and examples.	10	L2	CO2
OR					
Q.4	a.	Explain grep commands with all its options.	10	L2	CO2
	b.	Explain three standard files in UNIX.	06	L2	CO2
	c.	Explain the steps of shell interpretive cycle.	04	L2	CO2
Module – 3					
Q.5	a.	Explain POSIX and SUS (Single UNIX Specification) standards.	04	L2	CO3
	b.	Develop a C program to demonstrate the use of open() and read() system call in UNIX.	10	L3	CO3
	c.	Explain the use of mkdir() and rmdir() function in managing directories.	06	L2	CO3
OR					
Q.6	a.	Differentiate between character special files and block special files.	06	L2	CO3
	b.	Develop a c program to demonstrate the chdir() and fchdir() functions in UNIX.	10	L3	CO3
	c.	Explain the memory layout of a C program in UNIX.	04	L2	CO3
Module – 4					
Q.7	a.	Develop both the fork and vfork function in a example program.	10	L3	CO4
	b.	Explain briefly with an example two system v IPC mechanism: i) Message Queues ii) Semaphores	10	L2	CO4
OR					
Q.8	a.	Explain pipes and its limitations upon developing a program to send data from parent to child over a pipe.	10	L2	CO4
	b.	Explain the client server communication using FIFO with a neat diagram.	10	L2	CO4
Module – 5					
Q.9	a.	Illustrate signal in UNIX and develop program to setup signal handlers for sigsetjmp() and abort().	10	L3	CO5
	b.	Explain Daemon process by developing program to transform a normal user into a Daemon process.	10	L3	CO5
OR					
Q.10	a.	Explain implement SIGPROCMASK and SIGCONJMP functions with examples.	10	L2	CO5
	b.	Explain coding rules and error logging for Daemon process with neat diagram.	10	L2	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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-
1. Which of the following is a Producer?
 a) Animals b) Human beings c) Plants d) Fish
 2. An ecosystem consists of
 a) Biotic species b) Biotic species and physical factors
 c) Only insects d) Only materials
 3. The term Ecology was coined by
 a) E.P. Odium b) A. G. Tansley c) Albert Einstein d) A.G. Haeckel
 4. Which year did the concept of sustainability first appear?
 a) 1992 b) 1978 c) 1980 d) 1987
 5. According to United Nations, the following are the significant issues of social progress.
 a) Instruction b) Public Health c) Living standards d) All of these
 6. The number of SDGS are
 a) 10 b) 15 c) 17 d) 16
 7. Which one of the following is not comes under SDG's?
 a) Zero Hunger b) Political activity c) Education d) Gender quality
 8. Which of the following has the largest population in food chain?
 a) Producers b) 1st degree consumers
 c) 2nd degree consumers d) Decomposers
 9. U.N era of scheduling for sustainable growth is from
 a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14

10. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake
11. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these
12. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
13. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
14. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
15. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
16. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
17. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
18. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
19. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
20. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
21. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
22. The green house gas is
a) N₂O b) CH₄ c) CO₂ d) All of these
23. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
24. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
25. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB

26. What is the permissible range of p^H for drinking water as per the Indian standard?
 a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
27. Which of the following is a major cause of soil pollution?
 a) Accidents involving the vehicles that are transporting waste material.
 b) Pesticides and chemical fertilizers from agricultural lands
 c) Improper solid waste disposal d) All of these
28. What does E – Waste stand for?
 a) Environment Waste b) Electronic Waste
 c) Equipment Waste d) All of these
29. What is municipal solid waste disposal?
 a) The placement on the land b) The lack of waste
 c) In sanitary land fill site d) All of these
30. What is major cause of air pollution?
 a) Fossil fuel burning b) Industry emission
 c) Wild fire d) All of these
31. Average long – term weather of an area is called
 a) Weather conditions b) Seasonal variations
 c) Average weather d) Climate
32. Climate and global air circulations are mainly affected by the properties of
 a) Water and Air b) Temperature c) Precipitation d) None of these
33. The p^H of acid rain is
 a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2
34. Acid rain has been increasing day by day due to
 a) Urbanisation b) Industrialisation
 c) Increase in vehicle population d) None of these
35. Which is responsible for Ozone depletion?
 a) Methyl bromide b) CFC's
 c) Hydro chlorofluorocarbons d) All of these
36. The International protocol to protect the Ozone layer's
 a) Kytoto protocol b) Basal protocol c) Montreal protocol d) Viema protocol
37. The Ozone layer is locate upto _____ km above the earth surface.
 a) 1000 b) 50 c) 80 d) 100
38. Acid rain effects on
 a) Materials b) Plants c) Soil d) All of these
39. High – Concentration of fluoride content in drinking water causes.
 a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
40. Desired concentration of fluoride content in drinking water in mg/ℓ is
 a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75

41. The First United National conference as Human Environment was held at Stickholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
42. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
43. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
44. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
45. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
46. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
47. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
48. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
49. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
50. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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 4. Air pollution from automobiles can be controlled by providing
 - a) Wet collector
 - b) Scrubbers
 - c) Catalytic converter
 - d) All of these
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 - a) 120 dB
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19. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
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20. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron
21. Average long – term weather of an area is called
a) Weather conditions b) Seasonal variations
c) Average weather d) Climate
22. Climate and global air circulations are mainly affected by the properties of
a) Water and Air b) Temperature c) Precipitation d) None of these
23. The p^H of acid rain is
a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2

24. Acid rain has been increasing day by day due to
 - a) Urbanisation
 - b) Industrialisation
 - c) Increase in vehicle population
 - d) None of these
25. Which is responsible for Ozone depletion?
 - a) Methyl bromide
 - b) CFC's
 - c) Hydro chlorofluorocarbons
 - d) All of these
26. The International protocol to protect the Ozone layer's
 - a) Kytoto protocol
 - b) Basal protocol
 - c) Montreal protocol
 - d) Viema protocol
27. The Ozone layer is locate upto _____ km above the earth surface.
 - a) 1000
 - b) 50
 - c) 80
 - d) 100
28. Acid rain effects on
 - a) Materials
 - b) Plants
 - c) Soil
 - d) All of these
29. High – Concentration of fluoride content in drinking water causes.
 - a) Dental flurosis
 - b) Bone brittling
 - c) Dental caries
 - d) Plague
30. Desired concentration of fluoride content in drinking water in mg/ℓ is
 - a) 0.7 – 1.2
 - b) 0.5 – 2
 - c) 2 – 5
 - d) 1 – 1.75
31. Solar Energy is an example of
 - a) Renewable
 - b) Continuous
 - c) Non - renewable
 - d) None of these
32. Hydrogen energy can be tapped through
 - a) Heat pumps
 - b) Fuel cells
 - c) Petroleum
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33. OTEC is an energy technology that converts
 - a) Energy in large tides of ocean to generate electricity
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34. Which of the following source of energy is less eco – friendly?
 - a) Biogas
 - b) Wind
 - c) Solar
 - d) Nuclear
35. O₃ Represents
 - a) Oxygen
 - b) Planets
 - c) Ozone
 - d) Green house gases
36. Which of the following is an example of Water Borne Diseases
 - a) Hepatitis - A
 - b) Hepatitis - B
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37. Identify the renewable soured of energy from the following :
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38. Which of the following is biotic?
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39. World Environment Day celebrated on
 - a) 6th June
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40. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
41. Which of the following is a Producer?
a) Animals b) Human beings c) Plants d) Fish
42. An ecosystem consists of
a) Biotic species b) Biotic species and physical factors
c) Only insects d) Only materials
43. The term Ecology was coined by
a) E.P. Odiyum b) A. G. Tansley c) Albert Einstein d) A.G. Haeckal
44. Which year did the concept of sustainability first appear?
a) 1992 b) 1978 c) 1980 d) 1987
45. According to United Nations, the following are the significant issues of social progress.
a) Instruction b) Public Health c) Living standards d) All of these
46. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
47. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
48. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
49. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
50. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Average long – term weather of an area is called
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31. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5

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